Antenna Theory and Design

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Abstract

This report presents a structured study portfolio for *Antenna Theory and Design*, developed as part of graduate-level work in Electrical and Computer Engineering at Kennesaw State University. The material is organized chapter by chapter and focuses on the fundamental electromagnetic principles required to understand antenna behavior, including radiation mechanisms, Maxwell's equations, vector potential formulation, ideal dipoles, radiation patterns, directivity, gain, input impedance, radiation efficiency, and polarization.

The main objective of this work is not only to summarize antenna theory but also to connect the physical antenna layer to modern engineering applications, including radar systems, MIMO radar, beamforming, direction-of-arrival estimation, digital signal processing, and FPGA-based signal-processing architectures. Each chapter includes conceptual explanations, mathematical derivations, numerical examples, presentation materials, and solutions to educational problems. Emphasis is placed on understanding how antenna current distributions produce radiated fields and how antenna parameters affect received signal amplitude, phase, array manifold modeling, covariance-matrix formation, and ultimately radar and DOA estimation performance.

The report is based on independent study of Stutzman and Thiele's *Antenna Theory and Design*, 3rd Edition, together with original derivations, examples, figures, and explanatory notes prepared for educational and professional portfolio development. The long-term goal is to build a rigorous bridge between electromagnetic antenna theory and practical RF, radar, MIMO, DSP, and hardware-accelerated signal-processing systems.

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1.1 Summary of Chapter 2

Chapter 2 introduces the mathematical and physical foundations required for antenna analysis and design. While Chapter 1 provides a conceptual description of what antennas are and how they radiate, Chapter 2 develops the analytical framework needed to compute radiated fields and define the principal antenna parameters.

The central question of this chapter is:

> If the current distribution on an antenna is known, how can the radiated electromagnetic fields and antenna performance quantities be determined?

The fundamental chain of analysis is

$$\text{Antenna current} \longrightarrow \text{vector potential} \longrightarrow \text{radiated fields} \longrightarrow \text{antenna parameters.}$$

In symbolic form,

$$J \longrightarrow A \longrightarrow E, H \longrightarrow F(\theta, \phi), D, G, Z_A, e_r, \text{polarization.}$$

This chapter therefore builds the bridge between electromagnetic theory and practical antenna engineering.

1.1.1 Time-Harmonic Electromagnetic Fields

Antenna radiation problems are usually analyzed under sinusoidal steady-state excitation. For a time-harmonic electric field, the physical time-domain field may be written as

$$\mathcal{E}(\mathbf{r}, t) = \Re\left\{E(\mathbf{r})e^{j\omega t}\right\}.$$

The phasor $E(\mathbf{r})$ contains the spatial variation and complex phase information. With this convention, time derivatives are replaced by multiplication by $j\omega$. Thus, the time-harmonic form of Maxwell's equations becomes

$$\nabla \times E = -j\omega\mu H,$$

$$\nabla \times H = j\omega\epsilon E + J,$$

$$\nabla \cdot E = \frac{\rho}{\epsilon},$$

$$\nabla \cdot H = 0,$$

with the continuity equation

$$\nabla \cdot J = -j\omega\rho.$$

Here J is the impressed source current density on the antenna. In antenna theory, this current distribution is the physical source that produces the radiated fields.

The constitutive relations are

$$D = \epsilon E,$$

$$B = \mu H.$$

For most antenna radiation problems in air or free space,

$$\epsilon \approx \epsilon_0, \qquad \mu \approx \mu_0.$$

The intrinsic impedance of the medium is

$$\eta = \sqrt{\frac{\mu}{\epsilon}},$$

and in free space,

$$\eta_0 \approx 120\pi\ \Omega \approx 377\ \Omega.$$

1.1.2 Power Flow and the Poynting Vector

The flow of electromagnetic power is described by the complex Poynting vector:

$$S = \frac{1}{2} E \times H^*.$$

The real part of S represents time-average real power density in watts per square meter:

$$\Re\{S\} \quad [\mathrm{W/m^2}].$$

The average power flowing through a surface s is

$$P = \Re\left\{\iint_s S \cdot ds\right\} = \frac{1}{2}\Re\left\{\iint_s (E \times H^*) \cdot ds\right\}.$$

For an antenna, the total radiated power is found by integrating the outward radial power density over a closed surface in the far-field region.

This result is essential because it connects electromagnetic fields to measurable antenna quantities such as radiated power, radiation resistance, directivity, and gain.

1.1.3 Vector Potential Formulation

Solving Maxwell's equations directly for every antenna geometry is difficult. Therefore, the magnetic vector potential A is introduced. It is defined through

$$H = \frac{1}{\mu}\nabla \times A.$$

Using the Lorenz condition, the vector potential satisfies the wave equation

$$\nabla^2 A + \beta^2 A = -\mu J,$$

where

$$\beta = \omega\sqrt{\mu\epsilon}$$

is the phase constant. Since

$$\beta = \frac{2\pi}{\lambda},$$

the wavelength is related to frequency by

$$\lambda = \frac{c}{f},$$

where

$$c = \frac{1}{\sqrt{\mu_0\epsilon_0}} \approx 3 \times 10^8 \text{ m/s}.$$

For a general current distribution J, the vector potential is

$$A = \iiint_{v'} \mu J(\mathbf{r}')\frac{e^{-j\beta R}}{4\pi R}dv',$$

where R is the distance from the source point $\mathbf{r}'$ to the observation point.

The factor

$$\frac{e^{-j\beta R}}{4\pi R}$$

contains two important physical effects:

$$\frac{1}{R} \quad \text{amplitude spreading,}$$

and

$$e^{-j\beta R} \quad \text{phase delay due to propagation.}$$

Thus, the radiated field is the coherent sum of the contributions from all differential current elements on the antenna.

1.1.4 Far-Field Approximation

In the far field, the observation distance r is much larger than the dimensions of the antenna. The distance R from a source point to the observation point can be approximated as

$$R \approx r - \hat{r} \cdot \mathbf{r}'.$$

This gives the far-field vector potential

$$A = \mu \frac{e^{-j\beta r}}{4\pi r} \iiint_{v'} J(\mathbf{r}') e^{j\beta \hat{r} \cdot \mathbf{r}'} dv'.$$

Once A is known, the far electric field is obtained from the transverse components of the vector potential:

$$E = -j\omega(A_\theta \hat{\theta} + A_\phi \hat{\phi}).$$

The magnetic field is then found from the plane-wave relation

$$H = \frac{1}{\eta} \hat{r} \times E.$$

Therefore, in the far-field region,

$$E \perp H, \qquad E \perp \hat{r}, \qquad H \perp \hat{r},$$

and

$$\frac{|E|}{|H|} = \eta.$$

This means that the radiated field behaves locally like a plane wave, although globally it spreads outward as a spherical wave.

1.1.5 The Ideal Dipole

The ideal dipole is one of the most important elementary radiators in antenna theory. It is a very short current element of length Δz, where

$$\Delta z \ll \lambda.$$

The current is assumed to be uniform in magnitude and phase:

$$I(z) = I.$$

Although this is an idealized model, it is extremely useful because it reveals the basic physics of radiation.

For a z-directed ideal dipole located at the origin, the far-field components are

$$E_\theta = j\omega\mu \frac{I\Delta z}{4\pi r} e^{-j\beta r} \sin\theta,$$

$$H_\phi = j\beta \frac{I\Delta z}{4\pi r} e^{-j\beta r} \sin\theta.$$

The ratio of the fields is

$$\frac{E_\theta}{H_\phi} = \frac{\omega\mu}{\beta} = \sqrt{\frac{\mu}{\epsilon}} = \eta.$$

Thus, the far field of an ideal dipole has the structure of a transverse electromagnetic wave.

The angular dependence of the ideal dipole field is

$$F(\theta) = \sin\theta.$$

Therefore,

$$F(0^\circ) = 0,$$

$$F(90^\circ) = 1,$$

$$F(180^\circ) = 0.$$

The ideal dipole radiates maximum power perpendicular to its axis and no power along its axis. Its three-dimensional pattern has a doughnut shape.

1.1.6 Near Field and Far Field

The field surrounding an antenna can be divided into regions. Near the antenna, the fields are strongly influenced by stored electric and magnetic energy. Far away from the antenna, the fields behave as outward-propagating radiation.

The three important regions are:

$$\text{Reactive near field,}$$

$$\text{Radiating near field,}$$

$$\text{Far field.}$$

For electrically small antennas, the reactive near-field region extends approximately to

$$r \approx \frac{\lambda}{2\pi}.$$

In this region, stored energy dominates. The electric and magnetic fields exchange energy back and forth, similar to the energy exchange between a capacitor and an inductor.

In the far field, the radiated fields have the form

$$E, H \propto \frac{1}{r},$$

and the power density satisfies

$$S \propto \frac{1}{r^2}.$$

The radiation pattern becomes independent of distance and depends only on direction:

$$F = F(\theta, \phi).$$

For an antenna with maximum physical dimension D, a commonly used far-field distance is

$$r_{\mathrm{ff}} \approx \frac{2D^2}{\lambda}.$$

This condition is especially important in antenna measurement. If the antenna is measured too close to the source, the measured pattern may not represent the true far-field pattern.

1.1.7 Radiation Pattern

The radiation pattern describes the angular variation of the radiated field or power around an antenna. The normalized field pattern is defined as

$$F(\theta, \phi) = \frac{E(\theta, \phi)}{E_{\max}}.$$

The normalized power pattern is

$$P(\theta, \phi) = |F(\theta, \phi)|^2.$$

Antenna patterns are often plotted in decibels. For the field pattern,

$$F_{\text{dB}} = 20 \log_{10} |F|.$$

For the power pattern,

$$P_{\text{dB}} = 10 \log_{10} P = 10 \log_{10} |F|^2 = 20 \log_{10} |F|.$$

Therefore, the field pattern in dB and the power pattern in dB have the same numerical values.

Important radiation pattern features include:

main lobe,

side lobes,

back lobe,

nulls,

half-power beamwidth.

The half-power beamwidth, abbreviated HPBW, is the angular separation between the two directions where the power drops to one-half of its maximum value. Since power is proportional to field magnitude squared, the half-power points satisfy

$$|F| = \frac{1}{\sqrt{2}} \approx 0.707.$$

In decibels, these points correspond to

$$-3 \text{ dB}.$$

For the ideal dipole,

$$F(\theta) = \sin\theta.$$

The half-power points occur at

$$\theta = 45^\circ, \qquad \theta = 135^\circ.$$

Thus,

$$\text{HPBW} = 135^\circ - 45^\circ = 90^\circ.$$

1.1.8 Radiation from Line Currents

Many antennas can be modeled as line currents. For a z-directed line current $I(z')$, the far-field vector potential is

$$A_z = \mu \frac{e^{-j\beta r}}{4\pi r} \int I(z') e^{j\beta z' \cos\theta} dz'.$$

This equation shows that the radiation pattern is produced by the coherent summation of all current elements along the antenna.

The exponential term

$$e^{j\beta z' \cos\theta}$$

accounts for the phase difference between the contributions from different parts of the current distribution.

For a z-directed source, the far electric field is

$$E_\theta = j\omega A_z \sin\theta.$$

The corresponding normalized pattern can often be written as

$$F(\theta,\phi) = g(\theta,\phi)f(\theta,\phi),$$

where

$$g(\theta,\phi)$$

is the element factor, and

$$f(\theta,\phi)$$

is the pattern factor.

For a small z-directed current element,

$$g(\theta) = \sin\theta.$$

This concept is fundamental in array theory, where the total radiation pattern is written as

$$F_{\text{total}}(\theta,\phi) = F_{\text{element}}(\theta,\phi)F_{\text{array}}(\theta,\phi).$$

In radar, MIMO, beamforming, and DOA estimation, this relation is crucial because the array response is not determined only by element spacing. It is also affected by the individual element pattern.

1.1.9 Directivity

Directivity measures how strongly an antenna concentrates radiation in a particular direction compared with an isotropic radiator.

The radiation intensity is defined as

$$U(\theta, \phi) = r^2 S(\theta, \phi),$$

where S is the radial power density. Radiation intensity has units of watts per steradian.

The average radiation intensity is

$$U_{\text{ave}} = \frac{P_{\text{rad}}}{4\pi}.$$

The maximum directivity is

$$D = \frac{U_{\text{max}}}{U_{\text{ave}}}.$$

Equivalently,

$$D = \frac{4\pi U_{\text{max}}}{P_{\text{rad}}}.$$

The beam solid angle is defined as

$$\Omega_A = \iint_{4\pi} |F(\theta, \phi)|^2 d\Omega.$$

Then the directivity can be written as

$$D = \frac{4\pi}{\Omega_A}.$$

This expression shows that directivity is determined entirely by the shape of the radiation pattern. A narrower beam corresponds to a smaller beam solid angle and therefore a larger directivity.

For an ideal dipole,

$$F(\theta) = \sin\theta.$$

The beam solid angle is

$$\Omega_A = \int_0^{2\pi} \int_0^{\pi} \sin^2\theta \sin\theta \, d\theta d\phi = \frac{8\pi}{3}.$$

Therefore,

$$D = \frac{4\pi}{8\pi/3} = \frac{3}{2} = 1.5.$$

In decibels,

$$D_{\text{dBi}} = 10 \log_{10}(1.5) = 1.76 \text{ dBi}.$$

Thus, the ideal dipole has a directivity 50 percent larger than an isotropic radiator.

1.1.10 Gain

Directivity describes only the spatial concentration of radiated power. It does not include losses in the antenna. Gain includes the effect of radiation efficiency.

The radiation efficiency is

$$e_r = \frac{P_{\text{rad}}}{P_{\text{in}}},$$

where P_{rad} is the radiated power and P_{in} is the power accepted by the antenna.

The gain is

$$G = e_r D.$$

If the antenna is lossless,

$$e_r = 1,$$

and therefore

$$G = D.$$

For a lossy antenna,

$$e_r < 1,$$

and therefore

$$G < D.$$

In decibels,

$$G_{\mathrm{dB}} = 10 \log_{10} G,$$

and

$$D_{\mathrm{dB}} = 10 \log_{10} D.$$

Gain is usually expressed relative to an isotropic radiator and is then written in dBi. Gain may also be expressed relative to a half-wave dipole, in which case the unit is dBd.

Since a half-wave dipole has approximately

$$D \approx 1.64,$$

or

$$D \approx 2.15 \text{ dBi},$$

the conversion is

$$G(\mathrm{dBd}) = G(\mathrm{dBi}) - 2.15.$$

1.1.11 Antenna Impedance

From the viewpoint of the transmitter or receiver, an antenna behaves like an electrical load. The input impedance of an antenna is

$$Z_A = R_A + jX_A.$$

The real part R_A represents power dissipation. This dissipation has two components:

$$R_A = R_r + R_o,$$

where

$$R_r$$

is the radiation resistance, and

$$R_o$$

is the ohmic loss resistance.

The radiation resistance represents the part of the input resistance associated with power that is actually radiated away from the antenna. The ohmic resistance represents power converted into heat.

The average input power accepted by the antenna is

$$P_{\text{in}} = \frac{1}{2} R_A |I_A|^2.$$

The radiated power is

$$P_{\text{rad}} = \frac{1}{2} R_r |I_A|^2.$$

The ohmic loss power is

$$P_o = \frac{1}{2} R_o |I_A|^2.$$

Therefore,

$$P_{\text{in}} = P_{\text{rad}} + P_o.$$

The reactive part X_A is associated with energy stored in the near field of the antenna. A positive reactance corresponds to inductive behavior, while a negative reactance corresponds to capacitive behavior.

Antenna impedance is important because maximum power transfer requires impedance matching between the antenna and the connected transmission line. In many RF systems, the reference impedance is

$$Z_0 = 50\ \Omega.$$

If

$$Z_A \neq Z_0,$$

part of the incident power is reflected, reducing the delivered power.

1.1.12 Radiation Resistance of the Ideal Dipole

For an ideal dipole of length Δz, the radiated power is

$$P_{\text{rad}} = \frac{\eta\beta^2}{12\pi}(I\Delta z)^2.$$

The radiation resistance is defined by

$$R_r = \frac{2P_{\text{rad}}}{|I_A|^2}.$$

For the ideal dipole,

$$R_r = 80\pi^2 \left(\frac{\Delta z}{\lambda}\right)^2\ \Omega.$$

Since

$$\Delta z \ll \lambda,$$

the radiation resistance of an ideal dipole is very small.

For a practical short dipole, the current distribution is approximately triangular rather than uniform. The corresponding radiation resistance is

$$R_r = 20\pi^2 \left(\frac{\Delta z}{\lambda}\right)^2 \ \Omega.$$

This result has an important engineering implication:

Very small antennas usually have very small radiation resistance.

If the radiation resistance is small, even a modest ohmic resistance can severely reduce radiation efficiency.

1.1.13 Radiation Efficiency

Radiation efficiency is defined as

$$e_r = \frac{P_{\text{rad}}}{P_{\text{in}}}.$$

Using the resistance model,

$$e_r = \frac{R_r}{R_r + R_o}.$$

This equation shows that efficiency is high when

$$R_r \gg R_o,$$

and low when

$$R_r \ll R_o.$$

This is one of the main difficulties in designing electrically small antennas. Small antennas often have low radiation resistance and large reactance. As a result, they are difficult to match and may have poor efficiency.

1.1.14 Polarization

Polarization describes the time-varying orientation of the electric field vector of a radiated wave. It is determined by the curve traced by the tip of the instantaneous electric field vector at a fixed point in space.

The three principal polarization types are

$$\text{linear polarization,}$$

$$\text{circular polarization,}$$

and

$$\text{elliptical polarization.}$$

Consider a wave propagating in the $+z$-direction with transverse electric field

$$E = E_x\hat{x} + E_y\hat{y}.$$

The instantaneous components may be written as

$$E_x(t) = E_1\cos(\omega t),$$

$$E_y(t) = E_2\cos(\omega t + \delta).$$

If the two components are in phase or out of phase by 180°, the polarization is linear:

$$\delta = 0^\circ \quad \text{or} \quad \delta = 180^\circ.$$

If the amplitudes are equal and the phase difference is 90°, the polarization is circular:

$$E_1 = E_2, \qquad \delta = \pm 90^\circ.$$

The sign of δ determines the sense of rotation, commonly described as right-hand circular polarization or left-hand circular polarization.

In the most general case,

$$E_1 \neq E_2,$$

and

$$\delta \neq 0^\circ, 180^\circ,$$

so the polarization is elliptical.

Polarization is important because a receiving antenna responds most strongly to a wave with the same polarization. If the transmitting and receiving antennas are polarization mismatched, the received power is reduced.

For linearly polarized antennas, the polarization loss factor is

$$\mathrm{PLF} = \cos^2\psi,$$

where ψ is the angle between the transmit and receive polarization directions.

For example, if

$$\psi = 45^\circ,$$

then

$$\mathrm{PLF} = \cos^2(45^\circ) = 0.5.$$

In decibels,

$$10\log_{10}(0.5) = -3\ \mathrm{dB}.$$

Thus, a 45° polarization mismatch causes a 3 dB power loss.

1.1.15 Connection to Radar, MIMO, DOA, and DSP

The concepts of Chapter 2 are directly connected to radar, MIMO systems, beamforming, and direction-of-arrival estimation.

In radar systems, antenna gain directly affects the transmitted energy density in the direction of the target and the received echo strength. A higher gain antenna concentrates energy more effectively and improves the radar link budget.

In antenna arrays, the total radiation pattern can be expressed as

$$F_{\text{total}}(\theta, \phi) = F_{\text{element}}(\theta, \phi) F_{\text{array}}(\theta, \phi).$$

This relation is central to phased arrays, MIMO radar, and digital beamforming.

In DOA estimation algorithms such as MUSIC and ESPRIT, the array manifold is often modeled using ideal phase shifts between antenna elements. However, the true received signal is also affected by

$$\text{element pattern,}$$

$$\text{polarization,}$$

$$\text{mutual coupling,}$$

$$\text{impedance mismatch,}$$

and

$$\text{radiation efficiency.}$$

Therefore, antenna physics influences the covariance matrix used in signal processing:

$$\text{Antenna response} \longrightarrow \text{received signal} \longrightarrow \text{array covariance matrix} \longrightarrow \text{DOA estimate.}$$

For FPGA and DSP implementation, the algorithm may be digital, but the quality of the digital data depends on the analog electromagnetic front end. Poor antenna matching, polarization mismatch, distorted element patterns, or low efficiency can degrade the data before it reaches the ADC.

Thus, Chapter 2 is not only a theoretical foundation. It is also a practical foundation for radar system design, array processing, MIMO communications, and hardware-based signal processing.

1.1.16 Key Engineering Message of Chapter 2

The essential message of Chapter 2 is that an antenna must be understood through both its electromagnetic fields and its terminal behavior.

From the field viewpoint,

$$J \to A \to E, H \to F(\theta, \phi).$$

From the system viewpoint,

$$F(\theta, \phi), D, G, Z_A, e_r, \text{polarization}$$

determine how well the antenna performs in a real-world communication, radar, or sensing application.

Antenna performance cannot be described by a single number. A complete antenna the description requires the combined understanding of

1. radiation pattern,
2. directivity
3. gain,
4. input impedance,
5. radiation efficiency,
6. polarization.

In practical engineering terms:

An antenna is an electromagnetic transducer that converts guided electrical energy into radiated waves, and its quality is determined by how efficiently and directionally this conversion occurs.

This chapter provides the foundation for later topics such as short dipoles, half-wave dipoles, monopoles, arrays, aperture antennas, radar antennas, and antenna measurements.

1.2 Numerical Examples for Chapter 2: Antenna Fundamentals

This section presents numerical examples related to the main concepts of Chapter 2: the ideal dipole, radiation pattern, directivity and gain, antenna impedance, radiation efficiency, and polarization.

1.2.1 Example 1: Far-Field Radiation from an Ideal Dipole

Problem Statement: A z-directed ideal dipole operates in free space at $f = 300$ MHz. The dipole length is $\Delta z = \frac{\lambda}{50}$, and the peak current is $I = 1$ A. Find the far-zone electric field, magnetic field, power density, radiated power, and radiation resistance at $r = 100$ m, $\theta = 90°$. Assume free space: $\eta_0 = 120\pi\ \Omega \approx 377\ \Omega$, $c = 3 \times 10^8$ m/s.

Solution The wavelength is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{300 \times 10^6} = 1 \text{ m}.$$

Therefore,

$$\Delta z = \frac{\lambda}{50} = 0.02 \text{ m}.$$

The phase constant is

$$\beta = \frac{2\pi}{\lambda} = 2\pi \text{ rad/m}.$$

The far-field components of an ideal dipole are

$$E_\theta = j\omega\mu_0 \frac{I\Delta z}{4\pi r} e^{-j\beta r} \sin\theta,$$

$$H_\phi = j\beta \frac{I\Delta z}{4\pi r} e^{-j\beta r} \sin\theta.$$

Since

$$\omega\mu_0 = \eta_0\beta,$$

the magnitude of the electric field is

$$|E_\theta| = \eta_0\beta \frac{I\Delta z}{4\pi r} \sin\theta.$$

At $\theta = 90^\circ$,

$$\sin\theta = 1.$$

Thus,

$$|E_\theta| = 377(2\pi)\frac{(1)(0.02)}{4\pi(100)}.$$

Simplifying,

$$|E_\theta| = 377\frac{0.04\pi}{400\pi} = 377(10^{-4}) = 0.0377 \text{ V/m}.$$

The magnetic field magnitude is

$$|H_\phi| = \frac{|E_\theta|}{\eta_0} = \frac{0.0377}{377} = 1.0 \times 10^{-4} \text{ A/m}.$$

The time-average power density is

$$S_{\text{av}} = \frac{|E_\theta|^2}{2\eta_0}.$$

Therefore,

$$S_{\text{av}} = \frac{(0.0377)^2}{2(377)} = 1.89 \times 10^{-6}\ \text{W/m}^2.$$

The radiated power of an ideal dipole is

$$P_{\text{rad}} = \frac{\eta_0 \beta^2}{12\pi}(I\Delta z)^2.$$

Substituting values,

$$P_{\text{rad}} = \frac{377(2\pi)^2}{12\pi}(1 \times 0.02)^2.$$

$$P_{\text{rad}} = 0.158\ \text{W}.$$

The radiation resistance is

$$R_r = \frac{2P_{\text{rad}}}{|I|^2}.$$

Thus,

$$R_r = \frac{2(0.158)}{1^2} = 0.316\ \Omega.$$

Equivalently,

$$R_r = 80\pi^2 \left(\frac{\Delta z}{\lambda}\right)^2.$$

Since

$$\frac{\Delta z}{\lambda} = 0.02,$$

$$R_r = 80\pi^2(0.02)^2 = 0.316\ \Omega.$$

Numerical Results

$$\boxed{\lambda = 1\ \mathrm{m}}$$

$$\boxed{\Delta z = 0.02\ \mathrm{m}}$$

$$\boxed{|E_\theta| = 0.0377\ \mathrm{V/m}}$$

$$\boxed{|H_\phi| = 1.0 \times 10^{-4}\ \mathrm{A/m}}$$

$$\boxed{S_{\mathrm{av}} = 1.89 \times 10^{-6}\ \mathrm{W/m^2}}$$

$$\boxed{P_{\mathrm{rad}} = 0.158\ \mathrm{W}}$$

$$\boxed{R_r = 0.316\ \Omega}$$

Engineering Interpretation

Although the current is 1 A, the radiation resistance is only 0.316 Ω. This shows one of the main difficulties of electrically small antennas: they radiate, but their radiation resistance is very small. Therefore, even small ohmic losses can significantly reduce efficiency.

1.2.2 Example 2: Radiation Pattern and Half-Power Beamwidth of an Ideal Dipole

Problem Statement

An ideal z-directed dipole has normalized field pattern

$$F(\theta) = \sin\theta.$$

Find the normalized field pattern and power pattern at several angles, and determine the half-power beamwidth.

Solution

The normalized field pattern is

$$F(\theta) = \sin\theta.$$

The normalized power pattern is

$$P(\theta) = |F(\theta)|^2 = \sin^2\theta.$$

Evaluate the pattern at selected angles:

θ	$F(\theta) = \sin\theta$	$P(\theta) = \sin^2\theta$
$0°$	0	0
$30°$	0.5	0.25
$45°$	0.707	0.5
$60°$	0.866	0.75
$90°$	1	1
$120°$	0.866	0.75
$135°$	0.707	0.5
$150°$	0.5	0.25
$180°$	0	0

The half-power points occur where

$$P(\theta) = \frac{1}{2}.$$

Thus,

$$\sin^2\theta = \frac{1}{2}.$$

Therefore,

$$\sin\theta = \frac{1}{\sqrt{2}}.$$

The two half-power angles are

$$\theta_1 = 45^\circ, \qquad \theta_2 = 135^\circ.$$

Hence, the half-power beamwidth is

$$\text{HPBW} = \theta_2 - \theta_1 = 135^\circ - 45^\circ.$$

$$\boxed{\text{HPBW} = 90^\circ}$$

Engineering Interpretation

The ideal dipole has maximum radiation in the direction perpendicular to the dipole axis, that is, at

$$\theta = 90^\circ.$$

It has nulls along the dipole axis:

$$\theta = 0^\circ, \qquad \theta = 180^\circ.$$

The three-dimensional pattern is doughnut-shaped. This is a fundamental antenna pattern that appears repeatedly in dipoles, monopoles, and array elements.

1.2.3 Example 3: Directivity and Gain of an Ideal Dipole

Problem Statement

An ideal dipole has normalized field pattern

$$F(\theta) = \sin\theta.$$

Find its directivity. Then assume the radiation efficiency is

$$e_r = 0.85.$$

Find the gain in linear units and in dBi.

Solution

The beam solid angle is

$$\Omega_A = \int_0^{2\pi}\int_0^{\pi} |F(\theta)|^2 \sin\theta \, d\theta \, d\phi.$$

Since

$$F(\theta) = \sin\theta,$$

we have

$$\Omega_A = \int_0^{2\pi}\int_0^{\pi} \sin^2\theta \sin\theta \, d\theta \, d\phi.$$

$$\Omega_A = 2\pi \int_0^{\pi} \sin^3\theta \, d\theta.$$

Using

$$\int_0^{\pi} \sin^3\theta \, d\theta = \frac{4}{3},$$

we obtain

$$\Omega_A = 2\pi \left(\frac{4}{3}\right) = \frac{8\pi}{3}.$$

The directivity is

$$D = \frac{4\pi}{\Omega_A}.$$

Thus,

$$D = \frac{4\pi}{8\pi/3} = \frac{3}{2} = 1.5.$$

In dBi,

$$D_{\mathrm{dBi}} = 10\log_{10}(1.5) = 1.76 \text{ dBi}.$$

The gain is

$$G = e_r D.$$

Substituting

$$e_r = 0.85, \qquad D = 1.5,$$

gives

$$G = 0.85(1.5) = 1.275.$$

In dBi,

$$G_{\mathrm{dBi}} = 10\log_{10}(1.275) = 1.06 \text{ dBi}.$$

Final Answers

$$\boxed{D = 1.5}$$

$$\boxed{D = 1.76 \text{ dBi}}$$

$$\boxed{G = 1.275}$$

$$\boxed{G = 1.06 \text{ dBi}}$$

Engineering Interpretation

Directivity describes only the angular concentration of radiation. Gain includes radiation efficiency. Therefore, for a real antenna,

$$G < D.$$

This distinction is essential in radar and communication systems. A highly directive antenna with poor efficiency may still have poor realized performance.

1.2.4 Example 4: Antenna Impedance, Radiation Efficiency, and Mismatch

Problem Statement

A practical electrically short dipole operates at

$$f = 30 \text{ MHz}.$$

Its total length is

$$L = 0.5 \text{ m}.$$

Assume the antenna is a short dipole with approximately triangular current distribution. The ohmic loss resistance is

$$R_o = 0.20\ \Omega.$$

The input reactance is assumed to be capacitive:

$$X_A = -600\ \Omega.$$

Find:

1. the wavelength,
2. the radiation resistance,
3. the input resistance,
4. the radiation efficiency,
5. the approximate input impedance,
6. the reflection coefficient magnitude if connected directly to a 50 Ω line.

Solution

The wavelength is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{30 \times 10^6} = 10\ \text{m}.$$

The electrical length is

$$\frac{L}{\lambda} = \frac{0.5}{10} = 0.05.$$

For a practical short dipole,

$$R_r = 20\pi^2 \left(\frac{L}{\lambda}\right)^2.$$

Thus,

$$R_r = 20\pi^2(0.05)^2.$$

$$R_r = 0.494\ \Omega.$$

The input resistance is

$$R_A = R_r + R_o.$$

Therefore,

$$R_A = 0.494 + 0.20 = 0.694\ \Omega.$$

The radiation efficiency is

$$e_r = \frac{R_r}{R_r + R_o}.$$

$$e_r = \frac{0.494}{0.494 + 0.20} = 0.712.$$

Thus,

$$\boxed{e_r = 71.2\%}$$

The input impedance is

$$Z_A = R_A + jX_A.$$

Therefore,

$$\boxed{Z_A = 0.694 - j600\ \Omega}$$

If the antenna is connected directly to a 50 Ω transmission line, the reflection coefficient is

$$\Gamma = \frac{Z_A - Z_0}{Z_A + Z_0}.$$

With

$$Z_0 = 50\ \Omega,$$

$$\Gamma = \frac{(0.694 - j600) - 50}{(0.694 - j600) + 50}.$$

$$\Gamma = \frac{-49.306 - j600}{50.694 - j600}.$$

The magnitude is

$$|\Gamma| = \frac{\sqrt{(-49.306)^2 + (-600)^2}}{\sqrt{(50.694)^2 + (-600)^2}}.$$

$$|\Gamma| \approx 0.9998.$$

The fraction of incident power accepted by the antenna is

$$1 - |\Gamma|^2.$$

$$1 - |\Gamma|^2 \approx 3.83 \times 10^{-4}.$$

Therefore, only about

$$0.038\%$$

of the incident power is accepted without matching.

Final Answers

$$\boxed{\lambda = 10 \text{ m}}$$

$$\boxed{R_r = 0.494\ \Omega}$$

$$\boxed{R_A = 0.694\ \Omega}$$

$$\boxed{e_r = 0.712 = 71.2\%}$$

$$\boxed{Z_A = 0.694 - j600\ \Omega}$$

$$\boxed{|\Gamma| \approx 0.9998}$$

$$\boxed{1 - |\Gamma|^2 \approx 3.83 \times 10^{-4}}$$

Engineering Interpretation

This example illustrates an important practical point:

Efficiency and impedance matching are different issues.

The antenna has a radiation efficiency of about 71%, which is not terrible. However, because the input impedance is extremely different from 50 Ω, almost all of the incident power is reflected if the antenna is connected directly to a 50 Ω line.

Thus, electrically small antennas usually require a matching network. The matching network must both cancel the large reactance and transform the small resistance to the system impedance.

1.2.5 Example 5: Polarization Mismatch Between Two Linearly Polarized Antennas

Problem Statement

A transmitting antenna radiates a vertically polarized wave. A receiving antenna is linearly polarized but tilted by

$$\psi = 45^\circ$$

relative to the vertical direction.

Find the polarization loss factor and the loss in dB.

Solution

For two linearly polarized antennas, the polarization loss factor is

$$\mathrm{PLF} = \cos^2 \psi.$$

Substituting

$$\psi = 45^\circ,$$

we get

$$\mathrm{PLF} = \cos^2(45^\circ).$$

Since

$$\cos(45^\circ) = \frac{1}{\sqrt{2}},$$

$$\mathrm{PLF} = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}.$$

Thus,

$$\boxed{\mathrm{PLF} = 0.5}$$

The polarization mismatch loss in dB is

$$L_{\mathrm{pol,dB}} = 10\log_{10}(\mathrm{PLF}).$$

$$L_{\mathrm{pol,dB}} = 10\log_{10}(0.5) = -3.01 \text{ dB}.$$

Final Answers

$$\boxed{\mathrm{PLF} = 0.5}$$

$$\boxed{L_{\mathrm{pol}} = -3.01 \text{ dB}}$$

Engineering Interpretation

A 45° polarization mismatch causes a 3 dB power loss. This means that half of the available received power is lost only because of polarization misalignment.

This is important in wireless communication, radar, and antenna arrays. Even if the gain and impedance match are good, polarization mismatch can still reduce received power.

1.2.6 Example 6: Linear, Circular, and Elliptical Polarization

Problem Statement

A plane wave propagates in the $+z$-direction. Its transverse electric field components are

$$E_x(t) = E_1 \cos(\omega t),$$

$$E_y(t) = E_2 \cos(\omega t + \delta).$$

Classify the polarization for the following cases:

$$\text{Case A: } E_1 = 2 \text{ V/m}, \quad E_2 = 0, \quad \delta = 0°,$$

$$\text{Case B: } E_1 = 1 \text{ V/m}, \quad E_2 = 1 \text{ V/m}, \quad \delta = 90°,$$

$$\text{Case C: } E_1 = 2 \text{ V/m}, \quad E_2 = 1 \text{ V/m}, \quad \delta = 90°.$$

Solution

Case A The field has only one component:

$$E_x(t) = 2\cos(\omega t),$$

$$E_y(t) = 0.$$

The electric field oscillates along a fixed straight line. Therefore, the polarization is linear.

$$\boxed{\text{Case A: Linear polarization}}$$

Case B The two components have equal amplitudes:

$$E_1 = E_2 = 1 \text{ V/m},$$

and a phase difference of

$$\delta = 90°.$$

Equal orthogonal components with 90° phase difference produce circular polarization.

$$\boxed{\text{Case B: Circular polarization}}$$

Case C The two components are orthogonal and have a 90° phase difference, but their amplitudes are not equal:

$$E_1 = 2 \text{ V/m}, \qquad E_2 = 1 \text{ V/m}.$$

Therefore, the tip of the electric field vector traces an ellipse.

The axial ratio is

$$AR = \frac{E_{\text{major}}}{E_{\text{minor}}} = \frac{2}{1} = 2.$$

In dB,

$$AR_{\text{dB}} = 20 \log_{10}(2) = 6.02 \text{ dB}.$$

Thus,

$$\boxed{\text{Case C: Elliptical polarization}}$$

$$\boxed{AR = 2}$$

$$\boxed{AR_{\text{dB}} = 6.02 \text{ dB}}$$

Engineering Interpretation

Linear polarization is produced when the electric field remains along a fixed line. Circular polarization occurs when two equal orthogonal components differ in phase by 90°. Elliptical polarization is the general case.

In practical antenna systems, polarization must be included in the link budget. In radar and MIMO systems, polarization can also be used as an additional information channel or as a way to distinguish targets and propagation mechanisms.

1.2.7 Summary of Numerical Examples

The examples above illustrate the main engineering lessons of Chapter 2:

1. Ideal dipoles radiate with $F(\theta) = \sin\theta$.
2. Their maximum radiation occurs perpendicular to the dipole axis.
3. Their radiation resistance is very small when $\Delta z \ll \lambda$.
4. Directivity depends only on pattern shape.
5. Gain includes radiation efficiency.
6. Impedance matching and radiation efficiency are distinct, but both are important.
7. Polarization mismatch can cause significant loss of received power.

For radar, MIMO, DOA estimation, and beamforming, these ideas are not abstract. They directly affect the received signal amplitude, phase, covariance matrix, array manifold, and ultimately the performance of signal processing algorithms.

1.3 Problems

1.3.1 Section 2.1 Problems: Electromagnetic Fundamentals

Problem 1.3.1: 2.1-1

Paraphrased statement. Use the time-harmonic phasor definition to derive the phasor form of Faraday's law from the time-domain form.

Solution.

The time-domain Faraday law is

$$\nabla \times \mathcal{E} = -\frac{\partial \mathcal{B}}{\partial t}.$$

Using the phasor convention

$$\mathcal{E} = \Re\{E e^{j\omega t}\}, \qquad \mathcal{B} = \Re\{B e^{j\omega t}\},$$

we substitute into the time-domain equation:

$$\Re\{(\nabla \times E)e^{j\omega t}\} = -\Re\left\{\frac{\partial}{\partial t}\left(Be^{j\omega t}\right)\right\}.$$

Since the phasor B is independent of time,

$$\frac{\partial}{\partial t}\left(Be^{j\omega t}\right) = j\omega Be^{j\omega t}.$$

Therefore,

$$\Re\{(\nabla \times E)e^{j\omega t}\} = \Re\{(-j\omega B)e^{j\omega t}\}.$$

Equating phasors gives

$$\boxed{\nabla \times E = -j\omega B.}$$

Physical meaning. In phasor analysis, time differentiation becomes multiplication by $j\omega$. This is the electromagnetic analogue of AC circuit analysis.

Problem 1.3.2: 2.1-2

Paraphrased statement. Derive the divergence equation for E when conduction loss σ is included.

Solution.

The total current density is

$$J_T = \sigma E + J,$$

where J is the impressed source current. The total continuity equation is

$$\nabla \cdot J_T = -j\omega\rho_T.$$

Using $D = \epsilon E$, the total charge density is

$$\rho_T = \nabla \cdot D = \epsilon\nabla \cdot E.$$

The impressed current satisfies

$$\nabla \cdot J = -j\omega\rho.$$

Now substitute $J_T = \sigma E + J$ into the total continuity equation:

$$\nabla \cdot (\sigma E + J) = -j\omega\rho_T.$$

Assuming constant σ,

$$\sigma\nabla \cdot E + \nabla \cdot J = -j\omega\epsilon\nabla \cdot E.$$

Using $\nabla \cdot J = -j\omega\rho$,

$$\sigma\nabla \cdot E - j\omega\rho = -j\omega\epsilon\nabla \cdot E.$$

Thus,

$$(\sigma + j\omega\epsilon)\nabla \cdot E = j\omega\rho.$$

Therefore,

$$\nabla \cdot E = \frac{j\omega\rho}{\sigma + j\omega\epsilon}.$$

Equivalently, define the complex permittivity

$$\epsilon' = \epsilon - \frac{j\sigma}{\omega},$$

so that

$$j\omega\epsilon' = j\omega\epsilon + \sigma.$$

Then

$$\boxed{\nabla \cdot E = \frac{\rho}{\epsilon'}.}$$

For a lossless medium, $\sigma = 0$ and the result reduces to

$$\boxed{\nabla \cdot E = \frac{\rho}{\epsilon}.}$$

Physical meaning. Conduction loss can be absorbed into a complex permittivity. This is why lossy media can often be handled using the same field equations as lossless media, but with complex material parameters.

Problem 1.3.3: 2.1-3

Paraphrased statement. Assuming real ϵ and μ and no magnetic current, derive the complex Poynting theorem.

Solution.

Start from the phasor Maxwell curl equations:

$$\nabla \times E = -j\omega\mu H,$$

$$\nabla \times H = j\omega\epsilon E + J.$$

Taking the complex conjugate of the second equation gives

$$\nabla \times H^* = -j\omega\epsilon E^* + J^*.$$

Use the vector identity

$$\nabla \cdot (E \times H^*) = H^* \cdot (\nabla \times E) - E \cdot (\nabla \times H^*).$$

Substitute Maxwell's equations:

$$\nabla \cdot (E \times H^*) = H^* \cdot (-j\omega\mu H) - E \cdot (-j\omega\epsilon E^* + J^*).$$

Therefore,

$$\nabla \cdot (E \times H^*) = -j\omega\mu|H|^2 + j\omega\epsilon|E|^2 - E \cdot J^*.$$

Rearrange:

$$-E \cdot J^* = \nabla \cdot (E \times H^*) + j\omega\mu|H|^2 - j\omega\epsilon|E|^2.$$

Integrate over volume v and multiply by 1/2:

$$-\frac{1}{2}\iiint_v E \cdot J^*\,dv = \frac{1}{2}\iiint_v \nabla \cdot (E \times H^*)\,dv + \frac{j\omega}{2}\iiint_v \mu|H|^2\,dv - \frac{j\omega}{2}\iiint_v \epsilon|E|^2\,dv.$$

Using the divergence theorem,

$$\frac{1}{2}\iiint_v \nabla \cdot (E \times H^*)\,dv = \frac{1}{2}\iint_s (E \times H^*) \cdot ds.$$

Define

$$P_s = -\frac{1}{2}\iiint_v E \cdot J^*\,dv,$$

$$P_f = \frac{1}{2}\iint_s (E \times H^*) \cdot ds,$$

$$W_{m,av} = \frac{1}{4}\iiint_v \mu|H|^2\,dv,$$

$$W_{e,av} = \frac{1}{4}\iiint_v \epsilon|E|^2\,dv.$$

Then

$$\boxed{P_s = P_f + j2\omega(W_{m,av} - W_{e,av}).}$$

If conduction loss is included, with current σE, the dissipated average power is

$$P_{d,av} = \frac{1}{2}\iiint_v \sigma|E|^2\,dv,$$

so the full form is

$$\boxed{P_s = P_f + P_{d,av} + j2\omega(W_{m,av} - W_{e,av}).}$$

Physical meaning. The real part represents net power flow and ohmic loss. The imaginary part represents reactive energy exchange between electric and magnetic stored fields.

Problem 1.3.4: 2.1-4

Paraphrased statement. Write the complex power equation for a series RLC network in a form analogous to Poynting's theorem.

Solution.

For a series RLC circuit driven by a sinusoidal voltage source, let I be the peak phasor current. The impedance is

$$Z = R + j\left(\omega L - \frac{1}{\omega C}\right).$$

The complex power supplied by the source is

$$P_s = \frac{1}{2} V I^*.$$

Since $V = ZI$,

$$P_s = \frac{1}{2} Z |I|^2.$$

Thus,

$$P_s = \frac{1}{2} R |I|^2 + j\frac{1}{2}\left(\omega L - \frac{1}{\omega C}\right)|I|^2.$$

The average power dissipated in R is

$$P_R = \frac{1}{2} R |I|^2.$$

The average magnetic energy stored in the inductor is

$$W_{m,av} = \frac{1}{4} L |I|^2.$$

The capacitor voltage magnitude is

$$|V_C| = \frac{|I|}{\omega C},$$

so the average electric energy stored in the capacitor is

$$W_{e,av} = \frac{1}{4} C |V_C|^2 = \frac{|I|^2}{4\omega^2 C}.$$

Therefore,

$$j2\omega(W_{m,av} - W_{e,av}) = j\left[\frac{\omega L}{2}|I|^2 - \frac{1}{2\omega C}|I|^2\right].$$

So the circuit power theorem is

$$\boxed{P_s = P_R + j2\omega(W_{m,av} - W_{e,av}).}$$

Physical meaning. This is the circuit analogue of Poynting's theorem. The resistor dissipates real power, while the inductor and capacitor exchange reactive energy.

1.3.2 Section 2.2 Problems: Radiation Solutions of Maxwell's Equations

Problem 1.3.5: 2.2-1

Paraphrased statement. Derive the scalar wave equation for the electric scalar potential Φ.

Solution.

The electric field in terms of potentials is

$$E = -j\omega A - \nabla\Phi.$$

Take the divergence:

$$\nabla \cdot E = -j\omega\nabla \cdot A - \nabla^2\Phi.$$

Using the Lorenz condition,

$$\nabla \cdot A = -j\omega\mu\epsilon\Phi,$$

we get

$$\nabla \cdot E = -j\omega(-j\omega\mu\epsilon\Phi) - \nabla^2\Phi.$$

Since $(-j)(-j) = -1$,

$$\nabla \cdot E = -\omega^2\mu\epsilon\Phi - \nabla^2\Phi.$$

From Gauss's law,

$$\nabla \cdot E = \frac{\rho}{\epsilon}.$$

Therefore,

$$-\omega^2\mu\epsilon\Phi - \nabla^2\Phi = \frac{\rho}{\epsilon}.$$

Multiplying by -1 gives

$$\boxed{\nabla^2\Phi + \omega^2\mu\epsilon\Phi = -\frac{\rho}{\epsilon}.}$$

Since $\beta^2 = \omega^2\mu\epsilon$,

$$\boxed{\nabla^2\Phi + \beta^2\Phi = -\frac{\rho}{\epsilon}.}$$

Physical meaning. The scalar potential satisfies the same type of wave equation as the vector potential, but the source term is charge density rather than current density.

Problem 1.3.6: 2.2-2

Paraphrased statement. Use the phase expression of a wave to derive its propagation velocity.

Solution.

For a wave traveling in the $+z$ direction, the phase is

$$\psi = \omega t - \beta z.$$

A constant-phase point satisfies

$$\omega t - \beta z = \text{constant}.$$

Taking the time derivative,

$$\omega - \beta \frac{dz}{dt} = 0.$$

Thus,

$$\frac{dz}{dt} = \frac{\omega}{\beta}.$$

Therefore the phase velocity is

$$\boxed{v_p = \frac{\omega}{\beta}.}$$

Since

$$\beta = \omega\sqrt{\mu\epsilon},$$

we obtain

$$\boxed{v_p = \frac{1}{\sqrt{\mu\epsilon}}.}$$

In free space,

$$\boxed{v_p = c = \frac{1}{\sqrt{\mu_0\epsilon_0}} \approx 3 \times 10^8 \text{ m/s}.}$$

Physical meaning. A constant phase point moves at the phase velocity. In free space this is the speed of light.

Problem 1.3.7: 2.2-3

Paraphrased statement. For the scalar Green's function, show that $\psi = Ce^{-j\beta r}/r$ satisfies the homogeneous scalar wave equation away from the origin, and determine C.

Solution.

(a) Verification away from the origin. For a spherically symmetric function,

$$\nabla^2\psi = \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\psi}{dr}\right).$$

Let
$$\psi = C\frac{e^{-j\beta r}}{r}.$$
For $r \neq 0$, direct differentiation gives
$$\nabla^2\left(\frac{e^{-j\beta r}}{r}\right) = -\beta^2\frac{e^{-j\beta r}}{r}.$$
Thus,
$$\nabla^2\psi + \beta^2\psi = 0,$$
which proves that the assumed form satisfies the homogeneous equation for all $r \neq 0$.

(b) Determination of C. The point-source equation is
$$\nabla^2\psi + \beta^2\psi = -\delta(x)\delta(y)\delta(z).$$
Integrate over a small sphere of radius a centered at the origin:
$$\iiint_v \nabla^2\psi\, dv + \beta^2\iiint_v \psi\, dv = -1.$$
As $a \to 0$, the second integral tends to zero. By the divergence theorem,
$$\iiint_v \nabla^2\psi\, dv = \iint_s \nabla\psi \cdot ds.$$
Near the origin,
$$\psi \approx \frac{C}{r}, \qquad \frac{d\psi}{dr} \approx -\frac{C}{r^2}.$$
Therefore,
$$\iint_s \nabla\psi \cdot ds = \left(-\frac{C}{a^2}\right)(4\pi a^2) = -4\pi C.$$
Thus,
$$-4\pi C = -1,$$
so
$$\boxed{C = \frac{1}{4\pi}.}$$
Therefore,
$$\boxed{\psi = \frac{e^{-j\beta r}}{4\pi r}.}$$

Physical meaning. This is the outgoing spherical-wave Green's function. It is the fundamental building block used to form the radiation integral for arbitrary current distributions.

1.3.3 Section 2.3 Problems: The Ideal Dipole

Problem 1.3.8: 2.3-1

Paraphrased statement. Show that the compact electric-field expression for the ideal dipole follows from the expanded expression.

Solution.

Starting from the expanded electric field of the ideal dipole,

$$E_\theta = \frac{I\Delta z}{4\pi}\left[j\omega\mu\frac{1}{r} + \eta\frac{1}{r^2} + \frac{1}{j\omega\epsilon r^3}\right] e^{-j\beta r}\sin\theta,$$

$$E_r = \frac{I\Delta z}{2\pi}\left[\eta\frac{1}{r^2} + \frac{1}{j\omega\epsilon r^3}\right] e^{-j\beta r}\cos\theta.$$

Use

$$\beta = \omega\sqrt{\mu\epsilon}, \qquad \eta = \sqrt{\frac{\mu}{\epsilon}}, \qquad \frac{\omega\mu}{\beta} = \eta.$$

For the θ component, factor out

$$\frac{I\Delta z}{4\pi} j\omega\mu\frac{e^{-j\beta r}}{r}\sin\theta.$$

The bracket becomes

$$1 + \frac{\eta/r^2}{j\omega\mu/r} + \frac{1/(j\omega\epsilon r^3)}{j\omega\mu/r}.$$

Since

$$\frac{\eta}{j\omega\mu r} = \frac{1}{j\beta r},$$

and

$$\frac{1}{(j\omega\epsilon r^3)(j\omega\mu/r)} = -\frac{1}{\omega^2\mu\epsilon r^2} = -\frac{1}{(\beta r)^2},$$

we obtain

$$E_\theta = \frac{I\Delta z}{4\pi} j\omega\mu\left[1 + \frac{1}{j\beta r} - \frac{1}{(\beta r)^2}\right]\frac{e^{-j\beta r}}{r}\sin\theta.$$

For the radial component, factor out

$$\frac{I\Delta z}{2\pi}\eta\frac{e^{-j\beta r}}{r}\cos\theta.$$

Then

$$E_r = \frac{I\Delta z}{2\pi}\eta\left[\frac{1}{r} - j\frac{1}{\beta r^2}\right]\frac{e^{-j\beta r}}{r}\cos\theta.$$

Thus,

$$\boxed{E = E_\theta\hat{\theta} + E_r\hat{r}}$$

with the same form as the compact expression in the chapter.

Physical meaning. The terms proportional to $1/r$, $1/r^2$, and $1/r^3$ represent radiation, induction, and electrostatic near-field behavior, respectively.

Problem 1.3.9: 2.3-2

Paraphrased statement. Derive the complete electric field of an ideal dipole by two methods.

Solution.

Method (a): From the magnetic field. The ideal-dipole magnetic field is

$$H_\phi = \frac{I\Delta z}{4\pi} j\beta \left(1 + \frac{1}{j\beta r}\right) \frac{e^{-j\beta r}}{r} \sin\theta.$$

Away from the source,

$$E = \frac{1}{j\omega\epsilon} \nabla \times H.$$

For a field with only H_ϕ component,

$$(\nabla \times H)_r = \frac{1}{r\sin\theta} \frac{\partial}{\partial\theta}(\sin\theta H_\phi),$$

$$(\nabla \times H)_\theta = -\frac{1}{r}\frac{\partial}{\partial r}(rH_\phi).$$

Substituting H_ϕ into these two expressions gives

$$E_r = \frac{I\Delta z}{2\pi}\eta \left[\frac{1}{r} - j\frac{1}{\beta r^2}\right] \frac{e^{-j\beta r}}{r} \cos\theta,$$

$$E_\theta = \frac{I\Delta z}{4\pi} j\omega\mu \left[1 + \frac{1}{j\beta r} - \frac{1}{(\beta r)^2}\right] \frac{e^{-j\beta r}}{r} \sin\theta.$$

Method (b): From the vector potential. For a z-directed ideal dipole,

$$A = \hat{z}\,\mu I\Delta z \frac{e^{-j\beta r}}{4\pi r}.$$

The electric field is

$$E = -j\omega A - \frac{j}{\omega\mu\epsilon}\nabla(\nabla \cdot A).$$

Write

$$\hat{z} = \hat{r}\cos\theta - \hat{\theta}\sin\theta.$$

Then compute $\nabla \cdot A$ in spherical coordinates and substitute into the field expression. Retaining all terms gives the same result as above:

$$\boxed{E_\theta = \frac{I\Delta z}{4\pi} j\omega\mu \left[1 + \frac{1}{j\beta r} - \frac{1}{(\beta r)^2}\right] \frac{e^{-j\beta r}}{r} \sin\theta}$$

$$\boxed{E_r = \frac{I\Delta z}{2\pi}\eta\left[\frac{1}{r} - j\frac{1}{\beta r^2}\right]\frac{e^{-j\beta r}}{r}\cos\theta.}$$

Physical meaning. Both derivations demonstrate that the ideal dipole has a radiation field plus reactive near-field terms. The far field is transverse, but the near field also contains a radial electric component.

Problem 1.3.10: 2.3-3

Paraphrased statement. For the ideal dipole, find the complex Poynting vector in the general case, show that the time-average radiated power equals the standard result, and recover the near-field expression.

Solution.

Let

$$K = \frac{I\Delta z}{4\pi}, \qquad x = \beta r.$$

The general fields may be written as

$$H_\phi = K\, j\beta\left(1 + \frac{1}{jx}\right)\frac{e^{-j\beta r}}{r}\sin\theta,$$

$$E_\theta = K\, j\omega\mu\left(1 + \frac{1}{jx} - \frac{1}{x^2}\right)\frac{e^{-j\beta r}}{r}\sin\theta,$$

$$E_r = 2K\eta\left(\frac{1}{r} - \frac{j}{\beta r^2}\right)\frac{e^{-j\beta r}}{r}\cos\theta.$$

The complex Poynting vector is

$$S = \frac{1}{2}E \times H^*.$$

Since

$$E = E_r\hat{r} + E_\theta\hat{\theta}, \qquad H = H_\phi\hat{\phi},$$

we have

$$S = \frac{1}{2}E_\theta H_\phi^*\hat{r} - \frac{1}{2}E_r H_\phi^*\hat{\theta}.$$

Algebra gives

$$\boxed{S_r = \frac{|K|^2}{2r^2}\omega\mu\beta\left(1 - \frac{j}{(\beta r)^3}\right)\sin^2\theta}$$

$$\boxed{S_\theta = j\frac{|K|^2\eta\beta}{r^3}\left(1 + \frac{1}{(\beta r)^2}\right)\sin\theta\cos\theta.}$$

The real time-average outward power density is the real part of S_r:

$$\Re\{S_r\} = \frac{|K|^2}{2r^2}\omega\mu\beta\sin^2\theta.$$

The total time-average radiated power is

$$P = \int_0^{2\pi}\int_0^{\pi}\Re\{S_r\}r^2\sin\theta\,d\theta d\phi.$$

Thus,

$$P = \frac{|K|^2}{2}\omega\mu\beta(2\pi)\int_0^{\pi}\sin^3\theta\,d\theta.$$

Using

$$\int_0^{\pi}\sin^3\theta\,d\theta = \frac{4}{3},$$

we get

$$P = \frac{|K|^2}{2}\omega\mu\beta(2\pi)\frac{4}{3}.$$

Since $|K|^2 = |I\Delta z|^2/(16\pi^2)$,

$$\boxed{P = \frac{\omega\mu\beta}{12\pi}|I\Delta z|^2.}$$

This is the standard ideal-dipole radiated power.

In the near field, $\beta r \ll 1$, the dominant terms in S are

$$S_r^{nf} \approx -j\frac{\eta}{2\beta}\left(\frac{|I\Delta z|}{4\pi}\right)^2\frac{\sin^2\theta}{r^5},$$

$$S_\theta^{nf} \approx j\frac{\eta}{2\beta}\left(\frac{|I\Delta z|}{4\pi}\right)^2\frac{\sin 2\theta}{r^5}.$$

Thus,

$$\boxed{S^{nf} = -j\frac{\eta}{2\beta}\left(\frac{|I\Delta z|}{4\pi}\right)^2\frac{1}{r^5}\left(\sin^2\theta\,\hat{r} - \sin 2\theta\,\hat{\theta}\right).}$$

Physical meaning. The real radial component represents radiated power. The imaginary near-field terms represent reactive stored energy and do not produce net time-average power flow to infinity.

Problem 1.3.11: 2.3-4

Paraphrased statement. Verify that the ideal-dipole electric field satisfies $\nabla \cdot E = 0$ outside the source region.

Solution.

For the ideal dipole outside the current element, there is no charge density, so Maxwell's equation requires

$$\nabla \cdot E = 0.$$

The electric field has only r and θ components:

$$E = E_r \hat{r} + E_\theta \hat{\theta}.$$

In spherical coordinates,

$$\nabla \cdot E = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 E_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta E_\theta).$$

Substituting the complete expressions

$$E_r = \frac{I\Delta z}{2\pi}\eta \left[\frac{1}{r} - j\frac{1}{\beta r^2}\right]\frac{e^{-j\beta r}}{r}\cos\theta,$$

$$E_\theta = \frac{I\Delta z}{4\pi} j\omega\mu \left[1 + \frac{1}{j\beta r} - \frac{1}{(\beta r)^2}\right]\frac{e^{-j\beta r}}{r}\sin\theta,$$

and differentiating, the r-dependent terms cancel exactly with the θ-dependent terms. Hence,

$$\boxed{\nabla \cdot E = 0 \qquad (r \neq 0).}$$

Physical meaning. The divergence is zero outside the source because no free charge exists there. The field lines may curve and close, but they do not begin or end in the source-free region.

Problem 1.3.12: 2.3-5

Paraphrased statement. Show that the ratio of real radiated power density to reactive near-field power density is $(\beta r)^3$.

Solution.

At the direction of maximum radiation, $\theta = 90°$. The radiated power density from the ideal dipole is

$$S_{rad} = \frac{1}{2}\left(\frac{I\Delta z}{4\pi}\right)^2 \omega\mu\beta\frac{1}{r^2}.$$

The magnitude of the reactive near-field power density at $\theta = 90°$ is

$$|S_{nf}| = \frac{\eta}{2\beta}\left(\frac{I\Delta z}{4\pi}\right)^2 \frac{1}{r^5}.$$

Therefore,

$$\frac{S_{rad}}{|S_{nf}|} = \frac{\omega\mu\beta r^3}{\eta/\beta}.$$

Since

$$\eta = \frac{\omega\mu}{\beta},$$

we obtain

$$\boxed{\frac{S_{rad}}{|S_{nf}|} = (\beta r)^3.}$$

The two magnitudes are equal when

$$(\beta r)^3 = 1.$$

Thus,

$$\beta r = 1, \qquad r = \frac{1}{\beta} = \frac{\lambda}{2\pi}.$$

$$\boxed{r = \frac{\lambda}{2\pi}.}$$

Physical meaning. Inside the radiansphere, reactive stored energy dominates. Outside it, radiating power increasingly dominates.

1.3.4 Section 2.4 antproblems: Radiation Patterns and Field Regions

Problem 1.3.13: 2.4-1

Paraphrased statement. Derive the far-field electric field expression for a z-directed line source by retaining only $1/r$ terms.

Solution.

For a z-directed line source,

$$A = A_z\hat{z}.$$

In spherical coordinates,

$$\hat{z} = \hat{r}\cos\theta - \hat{\theta}\sin\theta.$$

Thus,

$$A = A_z\cos\theta\,\hat{r} - A_z\sin\theta\,\hat{\theta}.$$

The far-zone electric field is obtained from the transverse part of

$$E \approx -j\omega A.$$

Only the component transverse to $\hat{r}$ contributes to radiation. Therefore,

$$E_\theta = -j\omega(-A_z \sin\theta) = j\omega A_z \sin\theta.$$

Thus,

$$\boxed{E = j\omega A_z \sin\theta\,\hat{\theta}.}$$

Physical meaning. The radial component of A does not contribute to the far electric field. Radiation fields are transverse to the direction of propagation.

Problem 1.3.14: 2.4-2

Paraphrased statement. Derive the far-field path approximation $R \approx r - \hat{r}\cdot r'$.

Solution.

Let the observation vector be r and the source vector be r'. Then

$$R = |r - r'|.$$

Therefore,

$$R = [(r - r')\cdot(r - r')]^{1/2}.$$

Expanding,

$$R = [r^2 - 2r\cdot r' + r'^2]^{1/2}.$$

Factor out r^2:

$$R = r\left[1 - 2\frac{r\cdot r'}{r^2} + \frac{r'^2}{r^2}\right]^{1/2}.$$

In the far field, $r \gg r'$, so the term r'^2/r^2 is negligible. Then

$$R \approx r\left[1 - 2\frac{r\cdot r'}{r^2}\right]^{1/2}.$$

Using the binomial approximation

$$(1+x)^{1/2} \approx 1 + \frac{x}{2},$$

we obtain

$$R \approx r\left[1 - \frac{r\cdot r'}{r^2}\right].$$

Since $r = r\hat{r}$,

$$\frac{r \cdot r'}{r} = \hat{r} \cdot r'.$$

Thus,

$$\boxed{R \approx r - \hat{r} \cdot r'.}$$

Physical meaning. This approximation preserves the phase difference caused by source location while simplifying the amplitude factor. It is central to far-field radiation integrals and array phase calculations.

Problem 1.3.15: 2.4-3

Paraphrased statement. Show that the real and imaginary radial-dependence terms in the ideal-dipole fields become equal when $\beta r = 1$.

Solution.

For the magnetic field,

$$H_\phi \propto j\beta \left(1 + \frac{1}{j\beta r}\right) \frac{1}{r} = \left(\frac{j\beta}{r} + \frac{1}{r^2}\right).$$

The magnitudes of the two terms are equal when

$$\frac{\beta}{r} = \frac{1}{r^2},$$

which gives

$$\boxed{\beta r = 1.}$$

For the radial electric field,

$$E_r \propto \left(\frac{1}{r^2} - j\frac{1}{\beta r^3}\right).$$

The magnitudes are equal when

$$\frac{1}{r^2} = \frac{1}{\beta r^3},$$

so again

$$\boxed{\beta r = 1.}$$

For E_θ in the reactive near-field limit, neglect the radiation term:

$$E_\theta \propto j\omega\mu \left(\frac{1}{j\beta r} - \frac{1}{(\beta r)^2}\right) \frac{1}{r}.$$

The two remaining terms have equal magnitude when

$$\frac{1}{\beta r} = \frac{1}{(\beta r)^2},$$

which also yields

$$\boxed{\beta r = 1.}$$

Thus,

$$\boxed{r = \frac{1}{\beta} = \frac{\lambda}{2\pi}.}$$

Physical meaning. This distance marks the transition where reactive and radiative behavior have comparable strength for an ideal dipole.

Problem 1.3.16: 2.4-4

Paraphrased statement. Derive the ideal-dipole wave impedance E_θ/H_ϕ and evaluate its convergence toward η_0.

Solution.

The ideal-dipole fields are

$$E_\theta = Kj\omega\mu\left(1 + \frac{1}{j\beta r} - \frac{1}{(\beta r)^2}\right)\frac{e^{-j\beta r}}{r}\sin\theta,$$

$$H_\phi = Kj\beta\left(1 + \frac{1}{j\beta r}\right)\frac{e^{-j\beta r}}{r}\sin\theta.$$

Therefore,

$$Z_w(r) = \frac{E_\theta}{H_\phi} = \frac{\omega\mu}{\beta}\frac{1 + \frac{1}{j\beta r} - \frac{1}{(\beta r)^2}}{1 + \frac{1}{j\beta r}}.$$

Since

$$\frac{\omega\mu}{\beta} = \eta,$$

we get

$$\boxed{Z_w(r) = \eta\frac{1 + \frac{1}{j\beta r} - \frac{1}{(\beta r)^2}}{1 + \frac{1}{j\beta r}}.}$$

At

$$\beta r = 10,$$

$$\frac{Z_w}{\eta} \approx 0.9901 - j0.0010.$$

Thus,

$$\left|\frac{Z_w}{\eta}\right| \approx 0.9901, \qquad \angle Z_w \approx -0.057^\circ.$$

$$\boxed{Z_w \approx 0.990\eta \quad \text{at } \beta r = 10.}$$

For free space,

$$Z_w \approx 0.990(377) = 373\ \Omega.$$

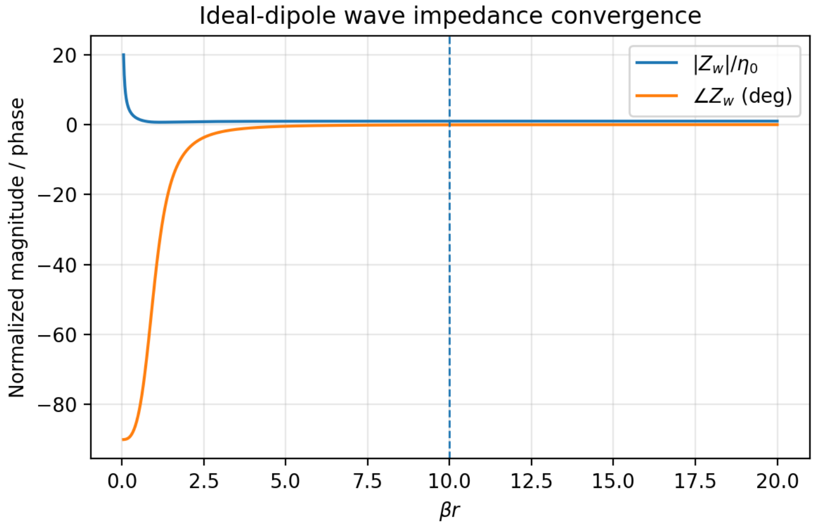

Figure 1.1: Convergence of the ideal-dipole wave impedance toward the intrinsic impedance.

Physical meaning. As the observation point moves into the far field, the wave impedance approaches the intrinsic impedance of the medium, confirming local plane-wave behavior.

Problem 1.3.17: 2.4-5

Paraphrased statement. Plot the magnitude error of E_θ relative to its far-field value and evaluate the error at $r = 5\lambda$.

Solution.

The exact E_θ correction factor relative to the far-field term is

$$C(x) = 1 + \frac{1}{jx} - \frac{1}{x^2}, \qquad x = \beta r.$$

The far-field approximation corresponds to $C = 1$. Therefore the relative error is

$$\epsilon(x) = |C(x) - 1| = \left| \frac{1}{jx} - \frac{1}{x^2} \right|.$$

Thus,

$$\epsilon(x) = \sqrt{\frac{1}{x^2} + \frac{1}{x^4}}.$$

In dB,

$$\epsilon_{dB} = 20 \log_{10} \epsilon(x).$$

For

$$r = 5\lambda,$$

$$x = \beta r = \frac{2\pi}{\lambda}(5\lambda) = 10\pi.$$

Therefore,

$$\epsilon = \sqrt{\frac{1}{(10\pi)^2} + \frac{1}{(10\pi)^4}} = 0.03185.$$

In dB,

$$\epsilon_{dB} = 20 \log_{10}(0.03185) = -29.94 \text{ dB}.$$

$$\boxed{\epsilon(5\lambda) \approx -30 \text{ dB}.}$$

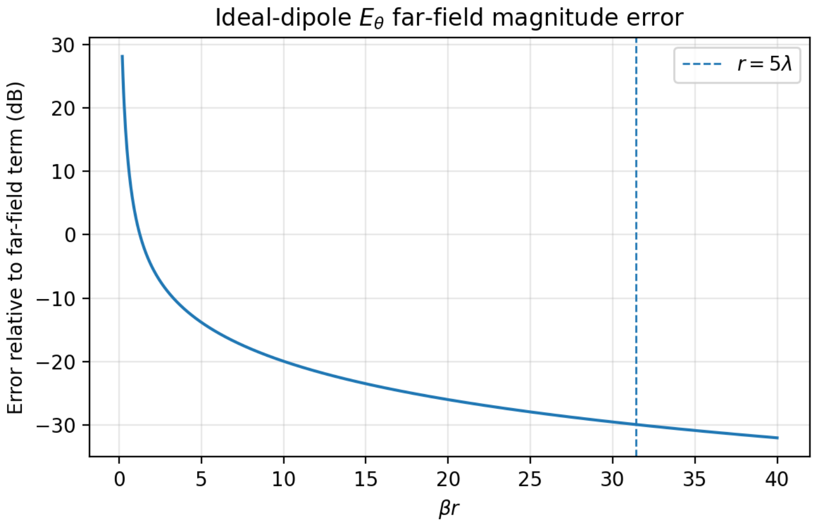

Figure 1.2: Magnitude error of the exact ideal-dipole E_θ relative to the far-field term.

Physical meaning. At $r = 5\lambda$, the non-radiating correction terms are about 30 dB below the radiation term, which supports the common far-field rule for small antennas.

Problem 1.3.18: 2.4-6

Paraphrased statement. Derive the reactive-radiating near-field boundary for a line source and find where it equals the ideal-dipole boundary.

Solution.

The fourth term in the expansion of R is proportional to

$$\frac{z'^3 \sin^2\theta\cos\theta}{2r^2}.$$

First maximize

$$f(\theta) = \sin^2\theta\cos\theta.$$

Differentiate:

$$\frac{df}{d\theta} = 2\sin\theta\cos^2\theta - \sin^3\theta = \sin\theta(2\cos^2\theta - \sin^2\theta).$$

The nonzero maximum satisfies

$$2\cos^2\theta = \sin^2\theta.$$

Using $\sin^2\theta = 1 - \cos^2\theta$,

$$2\cos^2\theta = 1 - \cos^2\theta,$$

$$3\cos^2\theta = 1.$$

Thus,

$$\cos\theta = \frac{1}{\sqrt{3}}, \qquad \boxed{\theta = 54.7^\circ.}$$

At this angle,

$$\sin^2\theta\cos\theta = \frac{2}{3\sqrt{3}}.$$

The maximum source coordinate is

$$z' = \frac{D}{2}.$$

Thus the maximum path error from this term is

$$\Delta R_{max} = \frac{(D/2)^3}{2r^2}\frac{2}{3\sqrt{3}} = \frac{D^3}{24\sqrt{3}r^2}.$$

Set this equal to $\lambda/16$, corresponding to a phase error of $\pi/8$:

$$\frac{D^3}{24\sqrt{3}r^2} = \frac{\lambda}{16}.$$

Solving for r,

$$r^2 = \frac{2D^3}{3\sqrt{3}\lambda}.$$

Hence,

$$\boxed{r = 0.62\sqrt{\frac{D^3}{\lambda}}.}$$

For equality with the ideal-dipole boundary,

$$0.62\sqrt{\frac{D^3}{\lambda}} = 0.16\lambda.$$

Let $x = D/\lambda$. Then

$$0.62x^{3/2} = 0.16.$$

Thus,

$$x = \left(\frac{0.16}{0.62}\right)^{2/3} = 0.405.$$

$$\boxed{D \approx 0.405\lambda.}$$

Physical meaning. For line sources larger than this, geometry-induced phase variation determines the boundary more strongly than the ideal-dipole radiansphere argument.

Problem 1.3.19: 2.4-7

Paraphrased statement. Show that $r = 5D$ produces a 10 percent maximum distance error for a line source of length D.

Solution.

The maximum distance difference from the center of the line source occurs at an end point:

$$R = r - \frac{D}{2}.$$

A 10 percent error means

$$R = 0.9r.$$

Therefore,

$$r - \frac{D}{2} = 0.9r.$$

Thus,

$$0.1r = \frac{D}{2},$$

so

$$\boxed{r = 5D.}$$

Physical meaning. This condition ensures that amplitude variations across the source are small enough for the far-field approximation.

Problem 1.3.20: 2.4-8

Paraphrased statement. Evaluate $r_{ff} = 2D^2/\lambda$ for several antenna lengths and determine whether the result is valid.

Solution.

The Rayleigh far-field criterion is

$$r_{ff} = \frac{2D^2}{\lambda}.$$

Case 1: $D = 5\lambda$**.**

$$r_{ff} = \frac{2(5\lambda)^2}{\lambda} = 50\lambda.$$

This is valid because it also satisfies $r > 5D = 25\lambda$ and $r > 1.6\lambda$.

$$\boxed{r_{ff} = 50\lambda \quad \text{valid.}}$$

Case 2: half-wave dipole, $D = \lambda/2$**.**

$$r_{ff} = \frac{2(\lambda/2)^2}{\lambda} = 0.5\lambda.$$

However,

$$5D = 2.5\lambda,$$

which is larger. Also 1.6λ is larger than 0.5λ. Thus 0.5λ is not sufficient.

$$\boxed{r_{ff} = 0.5\lambda \quad \text{from Rayleigh only, but not valid alone.}}$$

A safer value is at least

$$\boxed{r > 2.5\lambda.}$$

Case 3: short dipole, $D = 0.01\lambda$**.**

$$r_{ff} = \frac{2(0.01\lambda)^2}{\lambda} = 0.0002\lambda.$$

This is much too small. The criterion $r > 1.6\lambda$ is much larger, and for small antennas the more conservative rule $r \gtrsim 5\lambda$ is often used.

$$\boxed{r_{ff} = 0.0002\lambda \quad \text{not valid alone.}}$$

Physical meaning. The Rayleigh distance is dominant for electrically large antennas. For electrically small antennas, wavelength-based criteria dominate.

Problem 1.3.21: 2.4-9

Paraphrased statement. Make a graph of the far-field boundary criteria as functions of D/λ.

Solution.

Let

$$x = \frac{D}{\lambda}, \qquad \frac{r}{\lambda} = y.$$

The three criteria are

$$y_1 = 2x^2,$$

$$y_2 = 5x,$$

$$y_3 = 1.6.$$

The far-field region is above the largest of these three curves:

$$\boxed{\frac{r}{\lambda} > \max\left(2\left(\frac{D}{\lambda}\right)^2,\ 5\frac{D}{\lambda},\ 1.6\right).}$$

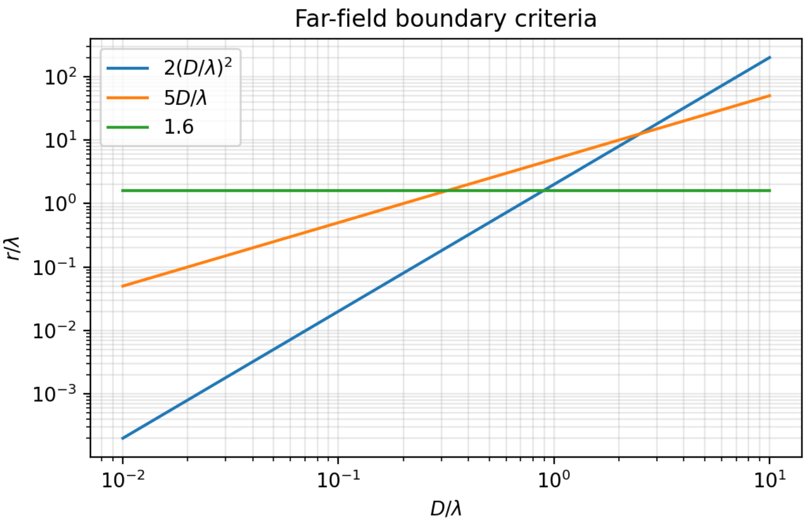

Figure 1.3: Far-field distance criteria versus normalized antenna size.

Physical meaning. The $2D^2/\lambda$ criterion controls large antennas. The 1.6λ criterion controls very small antennas. The $5D$ criterion is important in the intermediate region.

Problem 1.3.22: 2.4-10

Paraphrased statement. Determine the limiting far-field criterion for a 31-inch car-radio whip at 1 MHz.

Solution.

The antenna length is

$$D = 0.787 \text{ m}.$$

At

$$f = 1 \text{ MHz},$$

$$\lambda = \frac{3 \times 10^8}{10^6} = 300 \text{ m}.$$

Thus,

$$\frac{D}{\lambda} = \frac{0.787}{300} = 0.00262.$$

The three criteria are:

$$\frac{2D^2}{\lambda} = \frac{2(0.787)^2}{300} = 0.00413 \text{ m},$$

$$5D = 5(0.787) = 3.94 \text{ m},$$

$$1.6\lambda = 1.6(300) = 480 \text{ m}.$$

The largest is

$$\boxed{1.6\lambda = 480 \text{ m}.}$$

Therefore, the limiting criterion is the wavelength-based far-field condition.

Physical meaning. The car whip is electrically very small at 1 MHz, so its far-field boundary is not controlled by physical size but by wavelength.

1.3.5 Section 2.5 Problems: Directivity and Gain

Problem 1.3.23: 2.5-1

Paraphrased statement. Show that the total solid angle around a sphere is 4π steradians.

Solution.

The differential solid angle is

$$d\Omega = \sin\theta \, d\theta \, d\phi.$$

Integrating over all space,

$$\Omega = \int_0^{2\pi} \int_0^{\pi} \sin\theta \, d\theta d\phi.$$

The θ integral is

$$\int_0^{\pi} \sin\theta \, d\theta = 2.$$

The ϕ integral is

$$\int_0^{2\pi} d\phi = 2\pi.$$

Therefore,

$$\boxed{\Omega = 4\pi \text{ sr.}}$$

Physical meaning. The steradian is the angular analogue of area. A complete sphere contains 4π steradians.

Problem 1.3.24: 2.5-2

Paraphrased statement. A hemispherical power pattern is proportional to $|\cos^n\theta|$ for $0 \le \theta \le \pi/2$ and zero elsewhere. Compute directivity and HPBW for $n = 1, 2, 3$, sketch the patterns, and comment on $n = 0$.

Solution.

The normalized power pattern is

$$P(\theta) = \cos^n\theta, \qquad 0 \le \theta \le \frac{\pi}{2}.$$

The beam solid angle is

$$\Omega_A = \int_0^{2\pi} \int_0^{\pi/2} \cos^n\theta \sin\theta \, d\theta d\phi.$$

Let

$$u = \cos\theta, \qquad du = -\sin\theta \, d\theta.$$

Then

$$\Omega_A = 2\pi \int_0^1 u^n \, du = \frac{2\pi}{n+1}.$$

Thus,

$$D = \frac{4\pi}{\Omega_A} = 2(n+1).$$

So

$$\boxed{D = 2(n+1).}$$

For $n = 1, 2, 3$:

$$\boxed{D_1 = 4, \qquad D_2 = 6, \qquad D_3 = 8.}$$

The half-power point satisfies

$$\cos^n \theta_{HP} = \frac{1}{2}.$$

Therefore,

$$\theta_{HP} = \cos^{-1}\left(2^{-1/n}\right).$$

For a symmetric cut through the main beam, the HPBW is

$$\boxed{\mathrm{HPBW} = 2\cos^{-1}\left(2^{-1/n}\right).}$$

Numerically:

$$\boxed{\mathrm{HPBW}_{n=1} = 120^\circ,}$$

$$\boxed{\mathrm{HPBW}_{n=2} = 90^\circ,}$$

$$\boxed{\mathrm{HPBW}_{n=3} = 74.9^\circ.}$$

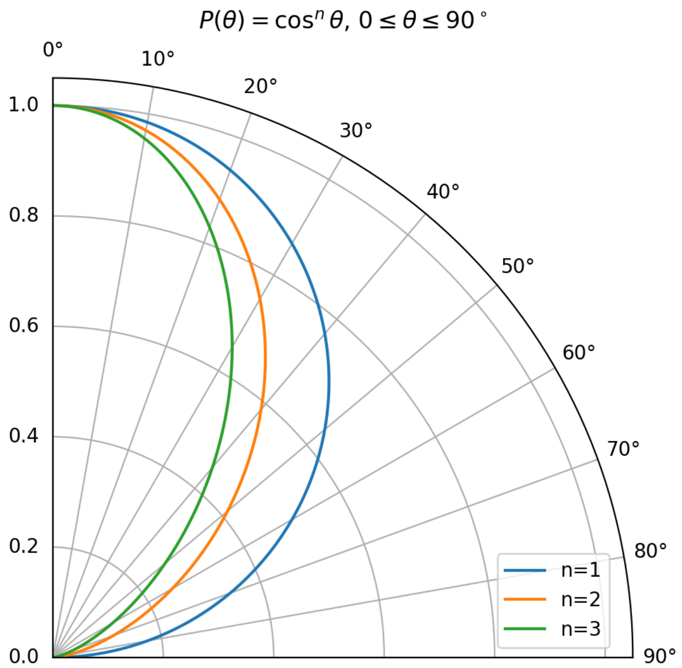

Figure 1.4: Power patterns $\cos^n\theta$ for $n = 1, 2, 3$.

For $n = 0$, the pattern is uniform over one hemisphere:

$$P(\theta) = 1, \qquad 0 \le \theta \le \frac{\pi}{2}.$$

Then

$$D = 2.$$

$$\boxed{D = 2 \quad \text{for uniform hemispherical radiation.}}$$

Physical meaning. Increasing n narrows the pattern and increases directivity. The case $n = 0$ radiates uniformly into one hemisphere, giving twice the directivity of an isotropic radiator.

Problem 1.3.25: 2.5-3

Paraphrased statement. Find the directivity of a piecewise field pattern independent of ϕ, and find the directivity at $\theta = 90°$.

Solution.

The field pattern is independent of ϕ and has the following values:

$$F = 1 \quad 0° \leq \theta \leq 30°,$$

$$F = 0.5 \quad 60° \leq \theta \leq 120°,$$

$$F = 0.707 \quad 150° \leq \theta \leq 180°,$$

with zero value elsewhere. The beam solid angle is

$$\Omega_A = 2\pi \sum_i |F_i|^2 \int_{\theta_{1i}}^{\theta_{2i}} \sin\theta \, d\theta.$$

Using

$$\int_{\theta_1}^{\theta_2} \sin\theta \, d\theta = \cos\theta_1 - \cos\theta_2,$$

we get

$$\Omega_A = 2\pi \left[(1)^2(1 - \cos 30°) + (0.5)^2(\cos 60° - \cos 120°) + (0.707)^2(\cos 150° - \cos 180°)\right].$$

Numerically,

$$\Omega_A = 2.833 \text{ sr}.$$

Thus,

$$D = \frac{4\pi}{\Omega_A} = 4.435.$$

$$\boxed{D = 4.44 \quad (6.47 \text{ dBi}).}$$

At $\theta = 90°$,

$$F = 0.5, \qquad |F|^2 = 0.25.$$

The directional directivity is

$$D(90°) = D|F(90°)|^2 = 4.435(0.25) = 1.109.$$

$$\boxed{D(90^\circ) = 1.11 \quad (0.45\ \text{dBi}).}$$

Physical meaning. Maximum directivity occurs in the strongest radiation region. Directional directivity elsewhere is reduced by the normalized power pattern.

Problem 1.3.26: 2.5-4

Paraphrased statement. Derive the approximate directivity formula using the two principal-plane half-power beamwidths.

Solution.

For a single-lobed pattern, the beam solid angle is approximated by

$$\Omega_A \approx \mathrm{HP}_E \mathrm{HP}_H,$$

where both beamwidths are in radians. Therefore,

$$D \approx \frac{4\pi}{\mathrm{HP}_E \mathrm{HP}_H}.$$

If the beamwidths are given in degrees,

$$\mathrm{HP}_{rad} = \mathrm{HP}_{deg} \frac{\pi}{180}.$$

Therefore,

$$D \approx \frac{4\pi}{\left(\mathrm{HP}_E^\circ \frac{\pi}{180}\right)\left(\mathrm{HP}_H^\circ \frac{\pi}{180}\right)}.$$

Thus,

$$D \approx \frac{4\pi (180)^2}{\pi^2 \mathrm{HP}_E^\circ \mathrm{HP}_H^\circ}.$$

The constant is

$$\frac{4(180)^2}{\pi} = 41253.$$

Hence,

$$\boxed{D \approx \frac{41253}{\mathrm{HP}_E^\circ \mathrm{HP}_H^\circ}.}$$

Physical meaning. This is a very useful engineering estimate: narrower beamwidths imply higher directivity.

Problem 1.3.27: 2.5-5

Paraphrased statement. Estimate the directivity of a low-side-lobe horn with equal half-power beamwidths of 29°.

Solution.

Using the approximate formula,

$$D \approx \frac{41253}{\mathrm{HP}_E^\circ \mathrm{HP}_H^\circ}.$$

Here

$$\mathrm{HP}_E^\circ = \mathrm{HP}_H^\circ = 29^\circ.$$

Therefore,

$$D \approx \frac{41253}{29(29)} = 49.05.$$

In decibels,

$$D_{dBi} = 10\log_{10}(49.05) = 16.91 \text{ dBi}.$$

$$\boxed{D \approx 49.1, \qquad D \approx 16.9 \text{ dBi}.}$$

Physical meaning. A horn with 29-degree beamwidth in both principal planes has moderate directivity, typical of aperture antennas with a relatively broad beam.

Problem 1.3.28: 2.5-6

Paraphrased statement. Derive the directivity of a sector pattern uniform over $\pi/2 - \alpha < \theta < \pi/2 + \alpha$ and zero elsewhere, and evaluate the result for $\alpha = \pi/6$.

Solution.

The field pattern is

$$F(\theta) = 1, \qquad \frac{\pi}{2} - \alpha < \theta < \frac{\pi}{2} + \alpha,$$

and zero elsewhere. The beam solid angle is

$$\Omega_A = \int_0^{2\pi} \int_{\pi/2-\alpha}^{\pi/2+\alpha} \sin\theta \, d\theta d\phi.$$

Thus,

$$\Omega_A = 2\pi[-\cos\theta]_{\pi/2-\alpha}^{\pi/2+\alpha}.$$

Since

$$\cos\left(\frac{\pi}{2} - \alpha\right) = \sin\alpha,$$

$$\cos\left(\frac{\pi}{2}+\alpha\right) = -\sin\alpha,$$

we obtain

$$\Omega_A = 2\pi(2\sin\alpha) = 4\pi\sin\alpha.$$

Therefore,

$$\boxed{D = \frac{4\pi}{\Omega_A} = \frac{1}{\sin\alpha}.}$$

For

$$\alpha = \frac{\pi}{6},$$

$$D = \frac{1}{\sin(\pi/6)} = 2.$$

$$\boxed{D = 2.}$$

Physical meaning. A narrower sector in elevation produces a larger directivity.

Problem 1.3.29: 2.5-7

Paraphrased statement. Derive the directivity of a cosecant pattern over a finite sector in θ and ϕ.

Solution.

The pattern is

$$F(\theta,\phi) = \csc\theta, \qquad \theta_1 < \theta < \frac{\pi}{2}, \quad 0 < \phi < \phi_0,$$

and zero elsewhere. The beam solid angle is

$$\Omega_A = \int_0^{\phi_0}\int_{\theta_1}^{\pi/2} |F(\theta,\phi)|^2 \sin\theta \, d\theta d\phi.$$

Since

$$|F|^2 = \csc^2\theta,$$

we get

$$\Omega_A = \phi_0 \int_{\theta_1}^{\pi/2} \csc^2\theta \sin\theta \, d\theta.$$

Thus,

$$\Omega_A = \phi_0 \int_{\theta_1}^{\pi/2} \csc\theta \, d\theta.$$

Using

$$\int \csc\theta \, d\theta = \ln\left|\tan\frac{\theta}{2}\right|,$$

we obtain

$$\Omega_A = \phi_0 \left[\ln \tan \frac{\theta}{2} \right]_{\theta_1}^{\pi/2}.$$

Since $\tan(\pi/4) = 1$,

$$\Omega_A = -\phi_0 \ln \left(\tan \frac{\theta_1}{2} \right).$$

Equivalently,

$$\Omega_A = \phi_0 \ln \left(\cot \frac{\theta_1}{2} \right).$$

Therefore,

$$\boxed{D = \frac{4\pi}{\phi_0 \ln \left(\cot \frac{\theta_1}{2} \right)}.}$$

Physical meaning. The cosecant pattern compensates for increasing range to the ground as the elevation angle decreases, which is useful in surface-search radar coverage.

Problem 1.3.30: 2.5-8

Paraphrased statement. Derive directivity expressions for a narrow Gaussian beam in terms of HPBW.

Solution.

The field pattern is modeled as

$$F(\theta) = \exp \left[-4 \ln \sqrt{2} \left(\frac{\theta}{\mathrm{HP}} \right)^2 \right] = \exp \left[-2 \ln 2 \left(\frac{\theta}{\mathrm{HP}} \right)^2 \right].$$

Thus the power pattern is

$$|F(\theta)|^2 = \exp \left[-4 \ln 2 \left(\frac{\theta}{\mathrm{HP}} \right)^2 \right].$$

For a narrow circularly symmetric beam,

$$\sin \theta \approx \theta,$$

and the upper limit can be extended to infinity:

$$\Omega_A \approx 2\pi \int_0^\infty \exp \left[-4 \ln 2 \left(\frac{\theta}{\mathrm{HP}} \right)^2 \right] \theta \, d\theta.$$

Use

$$\int_0^\infty e^{-a\theta^2} \theta \, d\theta = \frac{1}{2a}.$$

Here

$$a = \frac{4\ln 2}{\mathrm{HP}^2}.$$

Therefore,

$$\Omega_A \approx 2\pi \frac{1}{2a} = \frac{\pi}{a} = \frac{\pi \mathrm{HP}^2}{4\ln 2}.$$

The directivity is

$$D = \frac{4\pi}{\Omega_A} = \frac{16\ln 2}{\mathrm{HP}^2},$$

where HP is in radians. Thus,

$$\boxed{D \approx \frac{16\ln 2}{\mathrm{HP}_{rad}^2} = \frac{11.09}{\mathrm{HP}_{rad}^2}.}$$

If HP is in degrees,

$$\mathrm{HP}_{rad} = \mathrm{HP}^\circ \frac{\pi}{180}.$$

Thus,

$$D \approx \frac{16\ln 2}{(\mathrm{HP}^\circ \pi/180)^2}.$$

Numerically,

$$\boxed{D \approx \frac{36407}{(\mathrm{HP}^\circ)^2}.}$$

Physical meaning. Gaussian beams are widely used as practical models for aperture antennas, reflector antennas, and radar beams.

Problem 1.3.31: 2.5-9

Paraphrased statement. Compute the gain in dB for an antenna with directivity 20 and radiation efficiency 90 percent.

Solution.

Gain is

$$G = e_r D.$$

Given

$$e_r = 0.90, \qquad D = 20,$$

we get

$$G = 0.90(20) = 18.$$

In dB,

$$G_{dB} = 10\log_{10}(18) = 12.55 \text{ dB}.$$

$$\boxed{G = 18, \qquad G = 12.55 \text{ dBi.}}$$

Physical meaning. Efficiency reduces gain below directivity. A 90 percent efficient antenna loses about 0.46 dB relative to its directivity.

Problem 1.3.32: 2.5-10

Paraphrased statement. Compute the gain of an antenna with radiation efficiency 95 percent and a given piecewise field pattern.

Solution.

The field pattern is independent of ϕ:

$$F(\theta) = 1, \qquad 0^\circ \le \theta < 20^\circ,$$

$$F(\theta) = 0.707, \qquad 20^\circ \le \theta < 120^\circ,$$

$$F(\theta) = 0, \qquad 120^\circ \le \theta \le 180^\circ.$$

The beam solid angle is

$$\Omega_A = 2\pi \left[\int_0^{20^\circ} 1^2 \sin\theta \, d\theta + \int_{20^\circ}^{120^\circ} 0.707^2 \sin\theta \, d\theta \right].$$

Using numerical evaluation,

$$\Omega_A = 4.901 \text{ sr.}$$

Then

$$D = \frac{4\pi}{\Omega_A} = 2.564.$$

With radiation efficiency

$$e_r = 0.95,$$

the gain is

$$G = e_r D = 0.95(2.564) = 2.436.$$

In dB,

$$G_{dB} = 10 \log_{10}(2.436) = 3.87 \text{ dBi.}$$

$$\boxed{G = 2.44, \qquad G = 3.87 \text{ dBi.}}$$

Physical meaning. The gain is slightly less than directivity because the antenna has 5 percent loss.

1.3.6 Section 2.6 Problems: Antenna Impedance

Problem 1.3.33: 2.6-1

Paraphrased statement. Use the general ohmic resistance integral to verify the uniform-current and triangular-current ohmic resistance formulas.

Solution.

The general ohmic resistance expression is

$$R_o = \frac{1}{|I_A|^2}\frac{R_s}{2\pi a}\int_{-L/2}^{L/2} |I(z)|^2\, dz.$$

(a) Uniform current. For a uniform current,

$$I(z) = I_A, \qquad -\frac{L}{2} \le z \le \frac{L}{2}.$$

Thus,

$$\int_{-L/2}^{L/2} |I(z)|^2 dz = |I_A|^2 L.$$

Therefore,

$$R_o = \frac{1}{|I_A|^2}\frac{R_s}{2\pi a}|I_A|^2 L.$$

Hence,

$$\boxed{R_o = \frac{L}{2\pi a}R_s.}$$

(b) Triangular current. For a short dipole with triangular current,

$$I(z) = I_A\left(1 - \frac{2|z|}{L}\right), \qquad |z| \le \frac{L}{2}.$$

Then

$$\int_{-L/2}^{L/2} |I(z)|^2 dz = 2|I_A|^2 \int_0^{L/2}\left(1 - \frac{2z}{L}\right)^2 dz.$$

Let

$$u = 1 - \frac{2z}{L}, \qquad dz = -\frac{L}{2}du.$$

When $z = 0$, $u = 1$; when $z = L/2$, $u = 0$. Thus,

$$\int_0^{L/2}\left(1 - \frac{2z}{L}\right)^2 dz = \frac{L}{2}\int_0^1 u^2 du = \frac{L}{6}.$$

Therefore,

$$\int_{-L/2}^{L/2} |I(z)|^2 dz = 2|I_A|^2\frac{L}{6} = \frac{|I_A|^2 L}{3}.$$

Substitute into the ohmic resistance formula:

$$R_o = \frac{1}{|I_A|^2}\frac{R_s}{2\pi a}\frac{|I_A|^2 L}{3}.$$

Thus,

$$\boxed{R_o = \frac{L}{2\pi a}\frac{R_s}{3}.}$$

Physical meaning. Triangular current lowers ohmic loss compared with uniform current, but it lowers radiation resistance even more, which can reduce efficiency.

Problem 1.3.34: 2.6-2

Paraphrased statement. The textbook prints this as a second 2.6-1. For a tuned transmitting antenna, derive power delivered to radiation and ohmic losses, then simplify under resistance matching.

Solution.

Assume the antenna has

$$Z_A = R_r + R_o + jX_A.$$

Reactive tuning adds

$$X_L = -X_A,$$

so the net reactance is zero. The total series resistance is

$$R_T = R_g + R_r + R_o.$$

With generator peak voltage V_g, the current is

$$I = \frac{V_g}{R_g + R_r + R_o}.$$

The power delivered to radiation is

$$\boxed{P = \frac{1}{2}R_r|I|^2 = \frac{1}{2}R_r\frac{|V_g|^2}{(R_g + R_r + R_o)^2}.}$$

The ohmic loss in the antenna is

$$\boxed{P_o = \frac{1}{2}R_o|I|^2 = \frac{1}{2}R_o\frac{|V_g|^2}{(R_g + R_r + R_o)^2}.}$$

The power dissipated in the generator resistance is

$$\boxed{P_g = \frac{1}{2}R_g|I|^2 = \frac{1}{2}R_g\frac{|V_g|^2}{(R_g + R_r + R_o)^2}.}$$

Now assume resistance matching:

$$R_g = R_r + R_o.$$

Then

$$R_T = 2R_g.$$

The source current is

$$I = \frac{V_g}{2R_g}.$$

Thus,

$$P_g = \frac{1}{2}R_g\frac{|V_g|^2}{4R_g^2} = \frac{|V_g|^2}{8R_g}.$$

The total power supplied by the ideal voltage source and dissipated in all resistances is

$$P_{tot} = P_g + P + P_o = \frac{1}{2}(R_g + R_r + R_o)|I|^2.$$

Since $R_g = R_r + R_o$,

$$P_{tot} = \frac{1}{2}(2R_g)|I|^2 = R_g|I|^2.$$

Therefore,

$$\frac{P_g}{P_{tot}} = \frac{\frac{1}{2}R_g|I|^2}{R_g|I|^2} = \frac{1}{2}.$$

$$\boxed{\text{One-half of the supplied power is dissipated in } R_g.}$$

Physical meaning. For a conjugately matched source and load, maximum power transfer occurs, but half of the available source power is dissipated inside the source resistance.

1.3.7 Section 2.7 Problems: Radiation Efficiency

Problem 1.3.35: 2.7-1

Paraphrased statement. Compute the radiation efficiency of a 2-m aluminum dipole at 500 kHz for uniform and triangular current distributions.

Solution.

Given

$$L = 2 \text{ m}, \qquad f = 500 \text{ kHz}, \qquad d = 6.35 \text{ mm},$$
$$a = 3.175 \times 10^{-3} \text{ m}, \qquad \sigma_{Al} = 3.5 \times 10^{7} \text{ S/m}.$$

The wavelength is

$$\lambda = \frac{3 \times 10^8}{5 \times 10^5} = 600 \text{ m}.$$

The surface resistance is

$$R_s = \sqrt{\frac{\omega\mu_0}{2\sigma}}.$$

With $\omega = 2\pi f$,

$$R_s = 2.37 \times 10^{-4} \ \Omega.$$

(a) Uniform current. The radiation resistance is

$$R_r = 80\pi^2 \left(\frac{L}{\lambda}\right)^2.$$

Thus,

$$R_r = 80\pi^2 \left(\frac{2}{600}\right)^2 = 8.77 \times 10^{-3} \ \Omega.$$

The ohmic resistance is

$$R_o = \frac{L}{2\pi a} R_s.$$

Therefore,

$$R_o = \frac{2}{2\pi(3.175 \times 10^{-3})}(2.37 \times 10^{-4}) = 2.38 \times 10^{-2} \ \Omega.$$

Efficiency is

$$e_r = \frac{R_r}{R_r + R_o}.$$

Thus,

$$e_r = \frac{0.00877}{0.00877 + 0.02381} = 0.269.$$

$$\boxed{e_r = 26.9\% \quad \text{uniform current.}}$$

(b) Triangular current. For a short dipole with triangular current,

$$R_r = 20\pi^2 \left(\frac{L}{\lambda}\right)^2.$$

Thus,

$$R_r = 2.19 \times 10^{-3} \ \Omega.$$

The ohmic resistance is one third of the uniform-current value:

$$R_o = \frac{0.02381}{3} = 7.94 \times 10^{-3} \ \Omega.$$

Therefore,

$$e_r = \frac{0.00219}{0.00219 + 0.00794} = 0.217.$$

$$\boxed{e_r = 21.7\% \quad \text{triangular current.}}$$

Physical meaning. Although the triangular current reduces ohmic loss, it reduces radiation resistance by a larger factor, so efficiency is lower than in the uniform-current idealization.

Problem 1.3.36: 2.7-2

Paraphrased statement. Compute the radiation efficiency of a half-wavelength aluminum CB antenna at 27 MHz with radiation resistance 70 Ω.

Solution.

Given

$$f = 27 \text{ MHz}, \qquad R_r = 70\ \Omega, \qquad d = 6.35 \text{ mm},$$
$$a = 3.175 \times 10^{-3} \text{ m}, \qquad \sigma_{Al} = 3.5 \times 10^7 \text{ S/m}.$$

The wavelength is

$$\lambda = \frac{3 \times 10^8}{27 \times 10^6} = 11.11 \text{ m}.$$

A half-wavelength antenna has length

$$L = \frac{\lambda}{2} = 5.56 \text{ m}.$$

At 27 MHz,

$$R_s = 1.745 \times 10^{-3}\ \Omega.$$

Using the triangular-current approximation,

$$R_o = \frac{L}{2\pi a}\frac{R_s}{3}.$$

Thus,

$$R_o = \frac{5.56}{2\pi(3.175 \times 10^{-3})}\frac{1.745 \times 10^{-3}}{3} = 0.162\ \Omega.$$

The efficiency is

$$e_r = \frac{R_r}{R_r + R_o} = \frac{70}{70 + 0.162} = 0.9977.$$

$$\boxed{e_r = 99.77\%.}$$

Physical meaning. A resonant half-wave antenna has much larger radiation resistance than an electrically small antenna, so conductor loss is relatively negligible.

Problem 1.3.37: 2.7-3

Paraphrased statement. Compute the radiation efficiency of a 38-cm monopole at 50 MHz made from 4-mm-diameter aluminum tubing.

Solution.

Given

$$h = 0.38 \text{ m}, \qquad f = 50 \text{ MHz}, \qquad d = 4 \text{ mm},$$

$$a = 0.002 \text{ m}, \qquad \sigma_{Al} = 3.5 \times 10^7 \text{ S/m}.$$

The wavelength is

$$\lambda = \frac{3 \times 10^8}{50 \times 10^6} = 6 \text{ m}.$$

For a short monopole, using the monopole resistance as half of the corresponding dipole value,

$$R_r = 40\pi^2 \left(\frac{h}{\lambda}\right)^2.$$

Thus,

$$R_r = 40\pi^2 \left(\frac{0.38}{6}\right)^2 = 1.58 \ \Omega.$$

The surface resistance at 50 MHz is

$$R_s = 2.375 \times 10^{-3} \ \Omega.$$

The monopole ohmic resistance using the triangular-current approximation is

$$R_o = \frac{h}{2\pi a} \frac{R_s}{3}.$$

Thus,

$$R_o = \frac{0.38}{2\pi(0.002)} \frac{2.375 \times 10^{-3}}{3} = 0.0239 \ \Omega.$$

Efficiency is

$$e_r = \frac{R_r}{R_r + R_o} = \frac{1.58}{1.58 + 0.0239} = 0.985.$$

$$\boxed{e_r = 98.5\%.}$$

Physical meaning. At 50 MHz the monopole is not extremely small, and the radiation resistance is large enough to make conductor losses small.

Problem 1.3.38: 2.7-4

Paraphrased statement. Design a short dipole for a specified efficiency. Derive R_r/R_o, validate Example 2-4, and find the length for 90 percent efficiency using No. 18 copper wire at 100 MHz.

Solution.

(a) Derivation. For a short dipole of total length L,

$$R_r = 20\pi^2 \left(\frac{L}{\lambda}\right)^2.$$

The ohmic resistance is

$$R_o = \frac{L}{2\pi a}\frac{R_s}{3} = \frac{LR_s}{6\pi a}.$$

Thus,

$$\frac{R_r}{R_o} = \frac{20\pi^2(L/\lambda)^2}{LR_s/(6\pi a)}.$$

Therefore,

$$\boxed{\frac{R_r}{R_o} = \frac{120\pi^3 aL}{\lambda^2 R_s}.}$$

Let

$$r = \frac{R_r}{R_o}.$$

Then efficiency is

$$\boxed{e_r = \frac{r}{1+r}.}$$

(b) Validation using Example 2-4. The car whip example uses a short monopole. For a monopole of height h, the corresponding ratio is equivalent to using a dipole of length $2h$:

$$r = \frac{240\pi^3 ah}{\lambda^2 R_s}.$$

Using

$$h = 0.787 \text{ m}, \qquad a = 1.5875 \times 10^{-3} \text{ m},$$

$$\lambda = 300 \text{ m}, \qquad R_s = 1.405 \times 10^{-3}\ \Omega,$$

we obtain

$$r = 0.0732.$$

Thus,

$$e_r = \frac{0.0732}{1 + 0.0732} = 0.0682.$$

$$\boxed{e_r = 6.82\%,}$$

which agrees with Example 2-4.

(c) 90 percent efficiency for No. 18 copper wire at 100 MHz. For 90 percent efficiency,

$$e_r = 0.9 = \frac{r}{1+r}.$$

Thus,

$$r = 9.$$

For No. 18 copper wire,

$$d = 1.024 \text{ mm}, \qquad a = 0.512 \times 10^{-3} \text{ m}.$$

At

$$f = 100 \text{ MHz}, \qquad \lambda = 3 \text{ m}.$$

For copper,

$$\sigma = 5.8 \times 10^{7} \text{ S/m}.$$

The surface resistance is

$$R_s = \sqrt{\frac{\omega\mu_0}{2\sigma}} = 2.61 \times 10^{-3}\ \Omega.$$

Solve

$$r = \frac{120\pi^3 a L}{\lambda^2 R_s}$$

for L:

$$L = \frac{r\lambda^2 R_s}{120\pi^3 a}.$$

Substitute values:

$$L = \frac{9(3)^2(2.61 \times 10^{-3})}{120\pi^3(0.512 \times 10^{-3})} = 0.111 \text{ m}.$$

$$\boxed{L \approx 0.111 \text{ m} = 11.1 \text{ cm}.}$$

$$\boxed{L/\lambda \approx 0.037.}$$

Physical meaning. High efficiency is possible for a short dipole only if its conductor is sufficiently thick and long relative to wavelength, because radiation resistance must dominate ohmic resistance.

1.3.8 Section 2.8 Problems: Antenna Polarization

Problem 1.3.39: 2.8-1

Paraphrased statement. For an elliptically polarized wave with components $E_x = E_1 \cos(\omega t - \beta z)$ and $E_y = E_2 \cos(\omega t - \beta z + \delta)$, specify parameters for linear, RHCP, and LHCP polarization.

Solution.

The phase convention is

$$E_x = E_1 \cos(\omega t - \beta z),$$
$$E_y = E_2 \cos(\omega t - \beta z + \delta).$$

(a) Linear polarization with both components nonzero. Linear polarization occurs when the two components are in phase or 180° out of phase:

$$\boxed{E_1 \neq 0, \quad E_2 \neq 0, \quad \delta = 0^\circ \text{ or } 180^\circ.}$$

(b) Right-hand circular polarization. Circular polarization requires equal amplitudes and a 90° phase difference. With the convention used here, right-hand circular polarization corresponds to

$$\boxed{E_1 = E_2, \qquad \delta = -90^\circ.}$$

(c) Left-hand circular polarization. Left-hand circular polarization corresponds to

$$\boxed{E_1 = E_2, \qquad \delta = +90^\circ.}$$

Physical meaning. The sign of the phase difference determines the sense of rotation. Equal amplitudes give circular polarization; unequal amplitudes generally give elliptical polarization.

Problem 1.3.40: 2.8-2

Paraphrased statement. Write the frequency-domain vector electric fields for the polarizations in Problem 2.8-1.

Solution.

The phasor form is

$$E = E_1 \hat{x} + E_2 e^{j\delta} \hat{y}.$$

Linear polarization. If $\delta = 0^\circ$,

$$\boxed{E = E_1 \hat{x} + E_2 \hat{y}.}$$

If $\delta = 180°$,

$$\boxed{E = E_1\hat{x} - E_2\hat{y}.}$$

RHCP. For RHCP,

$$E_1 = E_2 = E_0, \qquad \delta = -90°.$$

Thus,

$$e^{j\delta} = e^{-j\pi/2} = -j.$$

Therefore,

$$\boxed{E = E_0(\hat{x} - j\hat{y}).}$$

LHCP. For LHCP,

$$E_1 = E_2 = E_0, \qquad \delta = +90°.$$

Thus,

$$e^{j\delta} = j.$$

Therefore,

$$\boxed{E = E_0(\hat{x} + j\hat{y}).}$$

Physical meaning. The complex unit vector compactly represents both the amplitude ratio and phase difference of the orthogonal field components.

Problem 1.3.41: 2.8-3

Paraphrased statement. Starting from $E = E_1\hat{x} + E_2 e^{j\delta}\hat{y}$, prove the normalized polarization-vector form.

Solution.

Start with

$$E = E_1\hat{x} + E_2 e^{j\delta}\hat{y}.$$

The squared magnitude is

$$|E|^2 = E \cdot E^*.$$

Since $\hat{x} \cdot \hat{y} = 0$,

$$|E|^2 = E_1^2 + E_2^2.$$

Thus,

$$|E| = \sqrt{E_1^2 + E_2^2}.$$

Define angle γ by

$$\cos\gamma = \frac{E_1}{|E|}, \qquad \sin\gamma = \frac{E_2}{|E|}.$$

Then

$$E_1 = |E| \cos\gamma, \qquad E_2 = |E| \sin\gamma.$$

Substitute into E:

$$E = |E| \cos\gamma\,\hat{x} + |E| \sin\gamma e^{j\delta}\hat{y}.$$

Therefore,

$$\boxed{E = |E| \left(\cos\gamma\,\hat{x} + \sin\gamma e^{j\delta}\hat{y}\right).}$$

Define the complex unit polarization vector

$$\hat{e} = \cos\gamma\,\hat{x} + \sin\gamma e^{j\delta}\hat{y}.$$

Then

$$\boxed{E = |E|\hat{e}.}$$

Also,

$$\hat{e} \cdot \hat{e}^* = \cos^2\gamma + \sin^2\gamma = 1.$$

Physical meaning. The pair (γ, δ) completely specifies the polarization state in terms of amplitude ratio and relative phase.

Problem 1.3.42: 2.8-4

Paraphrased statement. Prove that a RHCP wave normally incident on a perfect electric conductor reflects as LHCP.

Solution.

Let a RHCP wave travel in the $+z$ direction. Using the convention of this chapter, write the incident electric field phasor as

$$E_i = E_0(\hat{x} - j\hat{y})e^{-j\beta z}.$$

At a perfect electric conductor, the tangential electric field must vanish at the surface:

$$E_{tan,total} = E_i + E_r = 0.$$

Therefore the reflected electric field has reflection coefficient -1 for the tangential electric field:

$$E_r = -E_0(\hat{x} - j\hat{y})e^{+j\beta z}.$$

The exponential $e^{+j\beta z}$ indicates propagation in the $-z$ direction. The transverse vector is still proportional to

$$\hat{x} - j\hat{y},$$

but the sense of circular polarization is defined with respect to the direction of propagation. When the propagation direction reverses, the handedness reverses. Thus,

$$\boxed{\text{RHCP incident on a PEC reflects as LHCP.}}$$

Physical meaning. Circular polarization handedness is not just a property of the transverse vector; it is defined relative to the propagation direction. Reflection reverses the propagation direction, so the handedness changes.

Problem 1.3.43: 2.8-5

Paraphrased statement. Sketch a feed network that produces RHCP in the $+z$ direction using two orthogonal dipoles, and determine the polarization in the $-z$ direction.

Solution.

To generate circular polarization in the $+z$ direction, use two identical orthogonal dipoles:

$$\text{one along } x, \qquad \text{one along } y.$$

They must be fed with equal amplitudes and a 90° phase difference. For RHCP in the $+z$ direction using the convention of this chapter,

$$E_y \text{ must lag } E_x \text{ by } 90^\circ.$$

This can be achieved by adding an extra quarter-wavelength section of transmission line to the feed of the y-directed dipole:

$$\Delta\ell = \frac{\lambda}{4}.$$

The extra line length produces a phase delay

$$\Delta\phi = \beta\Delta\ell = \frac{2\pi}{\lambda}\frac{\lambda}{4} = \frac{\pi}{2} = 90^\circ.$$

Thus the feed condition is

$$I_y = I_x e^{-j90^\circ} = -jI_x.$$

A schematic description is:

source → equal splitter
branch 1: direct feed to x-dipole
branch 2: extra $\lambda/4$ line to y-dipole

The resulting field in the $+z$ direction is proportional to

$$E \propto \hat{x} - j\hat{y},$$

which is RHCP.

In the $-z$ direction, the propagation direction is reversed. The same rotating field is observed with opposite handedness relative to the new direction of propagation. Therefore,

$$\boxed{\text{Polarization in the } -z \text{ direction is LHCP.}}$$

Physical meaning. A simple quadrature feed can generate circular polarization, but the handedness reverses in the opposite radiation direction. This is important for dual-sided radiation and reflector/feed design.